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Underground pedestrian systems and citizens solar exposure risk during urban heatwaves: quantitative analysis and modeling☆

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ABSTRACT

Urban heatwaves pose significant risks to pedestrian comfort and mobility. To quantitatively evaluate the role of underground pedestrian systems (UPSs) in reducing pedestrians' sun-exposed heat risk during urban heatwaves, this study proposes a spatiotemporal exposure assessment framework based on a "path–risk" relationship. Taking the Tenjin district in Fukuoka, Japan, as a case study, the framework integrates meteorological observations, urban morphology, underground pedestrian flows, and railway passenger data to construct a three-dimensional pedestrian network. A multi-agent simulation is conducted to estimate pedestrians' potential solar exposure and heat risk under a hypothetical scenario in which the UPS is unavailable. The results demonstrate that the UPS can significantly reduce pedestrians' exposure to direct sunlight and heat risk during hot summer periods. The mitigation effect exhibits pronounced variations across intraday, weekly, seasonal, and interannual time scales and is jointly influenced by climatic conditions and travel demand. Spatially, the risk reduction benefits are unevenly distributed, with key entrances/exits and highly vulnerable street segments contributing disproportionately to the system's overall resilience. Temporal analyses further reveal that air temperature, sunshine duration, and pedestrian demand collectively shape the resilience response of the UPS. The proposed framework provides quantitative evidence for evaluating and optimizing the climate resilience performance of underground pedestrian systems under extreme heat conditions, while offering a transferable framework for long-term, multi-scale assessment of underground pedestrian networks under changing climatic conditions.

1. Introduction

Urban heatwaves, as one of the most pervasive climate extremes in recent years, have significantly altered the comfort of pedestrian microenvironments, indirectly affecting residents' travel behaviors and public health (Hess et al., 2023). During hot summer heatwaves, street environments exposed to direct solar radiation increase pedestrians' heat exposure risk (Basu et al., 2024; Melnikov et al., 2022; Azegami et al., 2023; Melnikov et al., 2017). Heat exposure reduces the comfort and safety of street-level walking environments, subsequently decreasing the willingness of residents to use public transportation such as subways and buses (Lin et al., 2022). This shift in travel behavior may lead to increased carbon emissions, thereby exacerbating urban heat island effects and contributing to global warming.

Underground pedestrian systems (UPSs) play a vital role in mitigating the impact of urban heatwaves. Characterized by stable environmental conditions, underground spaces provide effective protection against extreme weather. In recent years, UPSs in areas surrounding transit stations have evolved into large-scale, integrated networks, allowing pedestrians to directly access services within the station area (Gu et al, 2018; Wang et al, 2022). When surface-level pedestrian systems are disrupted by high temperatures or adverse weather, networked UPSs offer alternative routes for regional pedestrian mobility (Cui and Lin, 2016; Carmody and Sterling, 1983). Pedestrian flows in UPSs generate commercial vitality and enhance the provision of public services, further driving the development of such networks. As such, the planning of UPSs has become an essential strategy for balancing urban mobility demand and climate adaptation (Ahmed et al., 2024).

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Practical cases in Sapporo, Montreal, and Singapore have demonstrated that connected underground pedestrian networks can effectively mitigate the impact of snow accumulation (Zacharias, 2000) and solar exposure on transit passenger flows (Hess et al., 2023). However, UPSs exhibit different resilience performance under heatwave conditions, depending on whether they are semi-open or fully enclosed. Shaded, semi-open UPSs are effective at reducing heat stress driven by intense solar radiation (Thorsson et al., 2007). Their performance is limited in environments with high air temperatures and humidity. In such conditions, fully enclosed underground spaces equipped with indoor environmental control systems, such as air conditioning, are required to provide effective thermal protection. In the context of climate change and rising global temperatures, the extent to which reliance on the shading function of UPSs alone can reduce the heat exposure risk during urban heatwaves remains unclear. Establishing a measurable framework for evaluating UPS resilience performance is not only critical to guide decisions in underground space development (Broere & Mitew-Czajewska, 2026) but also to provide an evaluation tool for urban designers and planners seeking to enhance the adaptive capacity of UPS networks. However, there is a notable lack of systematic, quantitative assessments regarding the contribution of UPSs to pedestrian climate resilience in different districts. This study proposes a novel modeling and analytical framework to quantitatively evaluate the extent to which UPSs enhance climate resilience by mitigating pedestrians' exposure to direct solar radiation during urban heatwaves. The proposed framework provides performance-based indictors to support climate-adaptive planning and decision-making for urban infrastructure renewal.

Previous studies have proposed risk assessment frameworks based on the duration of interpersonal contact, which have been applied to quantify the effectiveness of crowd control strategies during epidemics (Gu et al., 2022a,b; Xiao et al., 2021). However, these frameworks primarily focus on pedestrian movement and risk dynamics within single-layer, enclosed pedestrian systems. As a result, these studies do not account for the potential role of pedestrian redistribution across multi-level pedestrian systems in reducing outdoor heat exposure. In areas with underground pedestrian networks, climate-related risk (extreme weather and heatwaves) and UPS users' movement are spatially separated across different levels of the pedestrian system (Zacharias & Wang, 2021). Thermal stress from extreme heat is primarily associated with ground-level pedestrian networks, where pedestrians are exposed to direct solar radiation and high temperatures. In contrast, users of UPSs, who are protected from direct climatic exposure, travel along alternative, underground routes physically separated from ground-level pedestrian networks. Consequently, the contribution of UPSs to mitigating the risk of heat exposure cannot be adequately captured through conventional path-based exposure assessments alone. To address this gap, this study proposes a hypothetical scenario–based framework to assess the contribution of UPSs to urban climate resilience (Wei et al., 2024). The resilience performance of UPS is quantified by simulating underground pedestrian trajectories and the resulting changes in heat exposure within ground-level pedestrian networks. Unlike conventional UPS evaluation studies that primarily assess the safety and operational efficiency of underground systems, the proposed framework focuses on quantifying the contribution of UPSs to reducing pedestrians' outdoor heat exposure. Because this mitigation effect represents the avoided exposure resulting from pedestrians' use of underground routes, it is inherently unobservable and therefore cannot be directly measured through field investigations alone.

Quantitatively evaluating the climatic shielding contribution of UPSs presents several challenges. The long-term climate resilience performance of UPSs cannot be adequately evaluated using a single scenario. In the context of summertime solar exposure, the efficacy of UPSs in reducing pedestrian exposure risk is influenced by a complex interplay of climatic conditions, street-level environmental factors, three-dimensional network structures, travel demand, and pedestrians' behavioral decisions. As such, the response of UPSs to heat stress represents a complex systems problem. From the perspective of environmental risk, the urban thermal environment is shaped by variables such as temperature, humidity, wind speed, and solar radiation (Huang et al., 2022), all of which exhibit significant spatial and temporal heterogeneity. Consequently, heat exposure scenarios dynamically vary across times of day, date, and weather conditions. In terms of pedestrian behavior, the underground pedestrian flows reflect distinct daily rhythms, causing the benefits of UPSs' climatic shielding to shift over time (Gu et al., 2018). Under the combined influence of these factors, the climatic shielding contribution of UPSs varies across multiple temporal scales. Furthermore, the unpredictable fluctuations in UPS use caused by urban public transport infrastructure development and public health emergencies, along with the prolonged recovery period, further amplify the uncertainty in the estimated contribution. While previous studies on heat exposure have largely overlooked the temporal dynamics and periodicity of influencing factors, this study investigates how variations in climatic conditions, urban morphology factors, and travel demand across multiple temporal scales jointly shape the dynamic resilience performance of UPSs.

The spatiotemporal mismatch between influencing factors represents a key source of complexity in evaluating the resilience performance of UPSs. Previous studies on UPS crowding and evacuation assessment generally assume that the spatiotemporal trends of crowd distribution and risk factors are aligned, allowing system risk to be evaluated using a limited number of representative adverse scenarios. Under urban heatwave conditions, however, the dynamic spatiotemporal distributions of climate, thermal environment, and travel activities make the path–risk relationship substantially more complex and uncertain. The spatial distribution of sunlight-exposed areas changes over time due to geographical factors such as solar position throughout the day and climatic regions. Moreover, urban morphology and vegetation cover create heterogeneous patterns of sunlit routes across the pedestrian network. Pedestrians' preference for shaded routes drives their adaptive route choice based on the distribution of sunlit areas (Lau et al., 2019). Understanding these dynamic spatial interactions is essential for evaluating potential thermal exposure and the mitigating role of UPSs. A further difficulty in this assessment is the temporal variation in travel demand. Origin-Destination (OD) matrices among UPS users change throughout the day. Differences in trip distributions between weekdays and weekends cause the path–risk relationship to fluctuate across the week. (Yang et al., 2026), which adds another dimension of spatiotemporal variability. In addition, ongoing regional development, the phased implementation of UPS construction, and climate change contribute to long-term shifts in travel demand and "path-risk" relationship. These dynamics result in two key outcomes:: first, the benefits offered by UPSs are not evenly distributed across different user groups; second, the operational efficiency of individual underground pedestrian facilities varies within the network. Therefore, examining the spatiotemporal distribution of user benefits and facility efficiency is important. Existing statistical approaches typically evaluate UPS performance using system-level indicators, such as total pedestrian volume and overall operational efficiency (Gu & Osaragi, 2026). However, these aggregate measures are insufficient to explain the dynamic spatial distribution of UPS performance across different user groups and urban locations. To address this limitation, this study adopts a spatially disaggregated analytical framework based on fine-grained spatial units to investigate how spatial mismatches in the path–risk relationship shape the spatiotemporal dynamics of UPS performance. Such insights provide valuable guidance for the planning, management, and operational maintenance of underground pedestrian spaces.

Compared with existing analytical and modeling frameworks, the proposed research presents three major methodological challenges. First, the diversity and temporal dynamics of influencing factors require simulations covering extended time periods with fine temporal resolution. Second, the diversity of evaluation targets and the heterogeneous spatiotemporal distribution of system performance require performance

indicators and explanatory models defined at the level of fine-grained travel-space units. Third, the heterogeneous distribution of resilience benefits across user groups requires analytical, evaluation, and visualization tools capable of capturing travel-activity characteristics. To address these challenges, this study proposes a climate resilience performance evaluation model for UPSs built upon a path–risk framework and spatiotemporal exposure simulation. Applying this model to the Tenjin district of Fukuoka, Japan, the study estimates the spatiotemporal distribution of solar exposure during the summer season and quantifies the contribution of the Tenjin UPS to reducing pedestrian solar exposure. Furthermore, spatial visualization metrics are developed to reveal the dynamic distribution of exposure risk, beneficiary populations, and facility response efficiency, thereby uncovering the mechanisms through which climatic conditions and travel demand affect the resilience performance of UPSs.

2. Methods

The basic framework of the model is illustrated in Fig. 1. This model integrates multi-source inputs, including meteorological observations, urban morphology, underground pedestrian flow data, and railway passenger data. The climate resilience contribution of UPSs is quantified by assessing the increase in travel-related heat exposure under a hypothetical scenario in which the UPS is inaccessible, based on the adaptive movement of underground users. It employs multi-agent simulation across various spatiotemporal scenarios to identify the spatial interaction between pedestrian routes and solar exposure risk. Based on the simulation results, the quantitative indicators are defined to measure the effectiveness of UPSs in mitigating pedestrians' solar heat exposure. By spatially coupling paths and exposure risks at the ground-link level, as well as Origin–Destination (OD) pairs and street blocks, distribution-based indicators are developed to visualize the spatiotemporal dynamics of the system's resilience response. The effects of environmental and behavioral factors are further examined through spatial visualization and statistical analyses. Based on the analysis results, the sensitivity of UPSs' resilience responses to fluctuations in influential factors, together with their resilience performance, is evaluated.

2.1. Study area and data processing

2.1.1. Study area

This study chooses the Tenjin district of Fukuoka City, Japan, as the case study area (Fig. 2(a)) for the simulation experiments. Fukuoka City is located at approximately 33°N latitude and features a humid subtropical climate with hot summers and mild winters. According to historical meteorological data published by the Japan Meteorological Agency, the highest average monthly temperatures typically occur in August. To capture the temporal coverage and representativeness of summer heat exposure, the simulation period is defined as spanning from July 1 to September 30.

This study selects the Tenjin underground shopping street and its surrounding interconnected urban blocks as the study area (Fig. 2(b)). The underground shopping street features a linear spatial layout and is well connected to adjacent urban blocks and transit stations, as illustrated in Fig. 3(a). Using open-access geospatial data obtained from OpenStreetMap (OSM), data on block morphology, building locations, and both aboveground and underground pedestrian pathways are extracted. Based on these data, a three-dimensional pedestrian network connecting facilities within the study area was constructed (Fig. 3(b)).

2.1.2. Available data

The model uses three primary data sources as input: meteorological observations, pedestrian flow counts, and streetview imagery.

Meteorological data are obtained from publicly available weather station records provided by Japan's Ministry of the Environment (JME) and the Japan Meteorological Agency (JMA). These datasets include hourly observations of temperature, humidity, weather conditions, wind speed, sunshine duration, and Wet-Bulb Globe Temperature (WBGT) from 2012 to 2024.

Pedestrian flow data are provided by the Fukuoka City Government and consist of bidirectional pedestrian counts collected at five-year intervals at major links of the Tenjin underground shopping street. These data were recorded hourly between 7:00 and 20:00 on both a weekday and a weekend. To represent the study period, the flow count data of 2012, 2017, and 2022 were used as input for estimating the Origin-Destination (OD) information of underground pedestrians.

Railroad usage data, consisting of the monthly passenger counts from three railway stations in the study area, were used to calibrate differences in monthly passenger demand between the data collection period and the simulation period. Google Street View imagery was collected via the Google Street View API at 10-m intervals along the street network, enabling detailed analyses of shading conditions and visual environmental features.

2.2. Risk scenario identification

A heatwave is typically defined as a weather event during which the maximum daily temperature exceeds 35 °C for at least three consecutive days. However, under direct solar radiation, pedestrians may experience comparable thermal stress at significantly lower ambient temperatures (Hess et al., 2023). To more accurately define heatwave conditions, this study adopts the Wet-Bulb Globe Temperature (WBGT) index, recommended by the Japan Ministry of the Environment (JME), as the key climatic parameter for heat risk assessment. According to the JME's heat alert policy, periods when the WBGT exceeds 31 (approximately equivalent to an air temperature of 35℃) are classified as "heatwave periods." Because direct solar radiation substantially increases pedestrians' thermal exposure, ground-level street segments receiving direct sunlight during WBGT-defined heatwave periods were considered as the main heat-risk locations in this study. When the WBGT in shaded outdoor environments exceeds 31, non-air-conditioned UPSs are unable to provide additional thermal protection beyond their solar-shading effect.

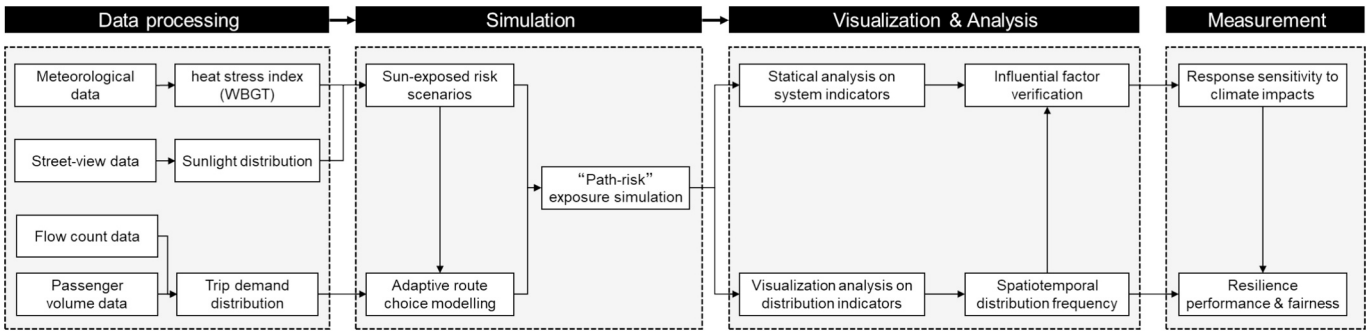


Fig. 1. Research Framework.

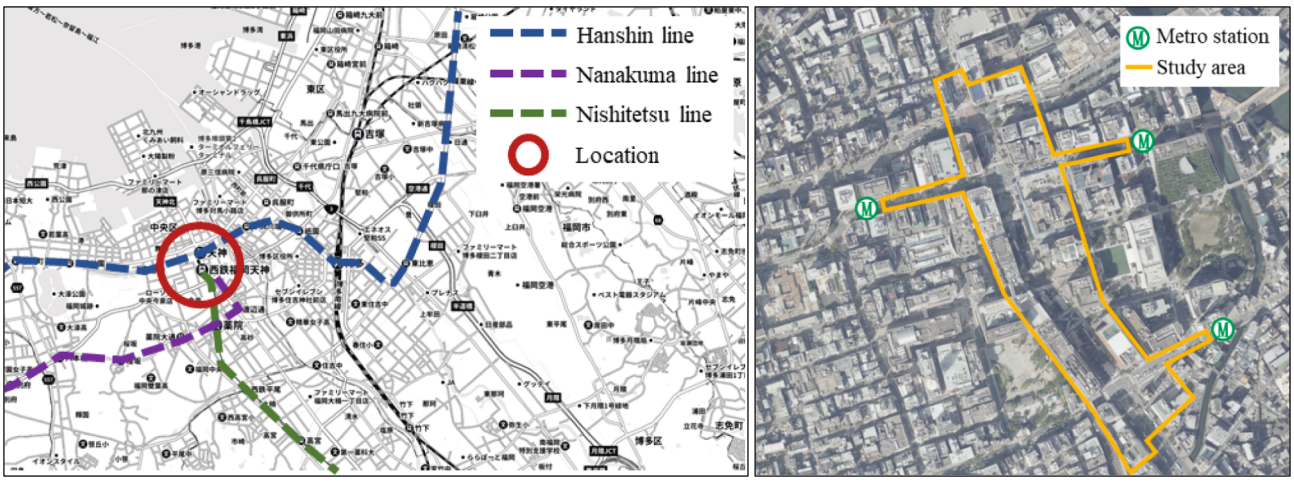


Fig. 2. Location of the study area in the Chuon district of Fukuoka, Japan: (a) geographic location; (b) study area.

Under such conditions, pedestrians can no longer effectively mitigate heat stress simply by using underground routes. Because this study focuses specifically on the contribution of UPSs' solar-shading function to reducing heat exposure under high-risk conditions, this situation is considered beyond the scope of the present resilience assessment framework. Therefore, the spatiotemporal distribution of these heat-risk street segments is therefore jointly determined by the timing of extreme heat and the urban morphology governing solar exposure on ground-level streets.

To assess whether the UPS can alleviate heat exposure on ground-level streets, this study estimates WBGT values under shaded conditions. Using weather station observations and the heat stress index framework provided by the JME and JMA, the WBGT values for each street link across different time periods under both sunlit and shaded scenarios are shown in Fig. 4. The results indicate that heat exposure can be substantially reduced when pedestrians travel through the underground pedestrian network, thereby avoiding direct solar radiation.

The spatiotemporal distribution of sun-exposed street segments is determined by urban morphology, street orientation, and street-level environmental conditions. To capture the spatiotemporal dynamics of solar exposure across the pedestrian network, this study adopts a sequential Google Street View image analysis approach (Zhu & Gu, 2022). As illustrated in Fig. 5, temporal information on sunlit conditions at each sampling location (Fig. 5(a)) was aggregated to derive time-series data for the sunlit and shaded lengths of each ground-level street segment. These data were subsequently integrated with WBGT values to generate the spatiotemporal distribution of sun-exposure risk across the pedestrian network.

2.3. Sunlit exposure simulation

2.3.1. OD matrix estimation

Based on the pedestrian survey data, a set of hour-specific OD matrices was obtained using the entropy maximization method (Fig. 6 (a)). A detailed description of the OD estimation procedure is provided in Gu and Osaragi (2016). The estimated OD matrix, $T_{ij,\,u}$ represents the number of trips between each origin–destination (OD) pair i and j during the u-th hour and was used as the simulation input (Fig. 6(b)).

Using the pedestrian flow counts, hour-specific OD matrices for both weekdays and weekends were estimated for 2012, 2017, and 2022. Given the limited availability of continuous OD matrices, this study adopts a widely used approach to reconstructing long-term travel demand, rooted in the established correlation between metro passenger volume and pedestrian flow counts (Zacharias & He, 2018). Specifically, we assume that the trip-to-passenger ratio for each OD pair remains stable within a given year and evolves gradually across consecutive observation years. To operationalize this assumption for unobserved periods, we apply linear interpolation to estimate the missing annual ratios (Gu & Osaragi, 2026). In addition, to account for seasonal variations in travel behavior, we calibrate the estimated OD matrices using monthly passenger count ratios, adjusting from the data collection month to the target simulation period. The validity of the estimated results is expected to improve when adopting flow-count data that align with the study period.

2.3.2. Route choice simulation

The route choice model considers pedestrian preference for shaded routes during extreme heat events. A utility-based cost function was constructed for each street link, incorporating factors such as route length, sun-exposure duration, intersection waiting time, and vertical movement. The cost function is formulated as follows:

$$C_{lt} = d_l + \alpha_1 S_l + \alpha_2 W_l + \alpha_3 D_{lt} \tag{1}$$

The coefficient values (α_n) were determined based on previous studies. Detailed references and parameter settings are provided in Table 1. All weighting coefficients represent equivalent distance conversion factors used to transform different impedance factors into generalized link costs.

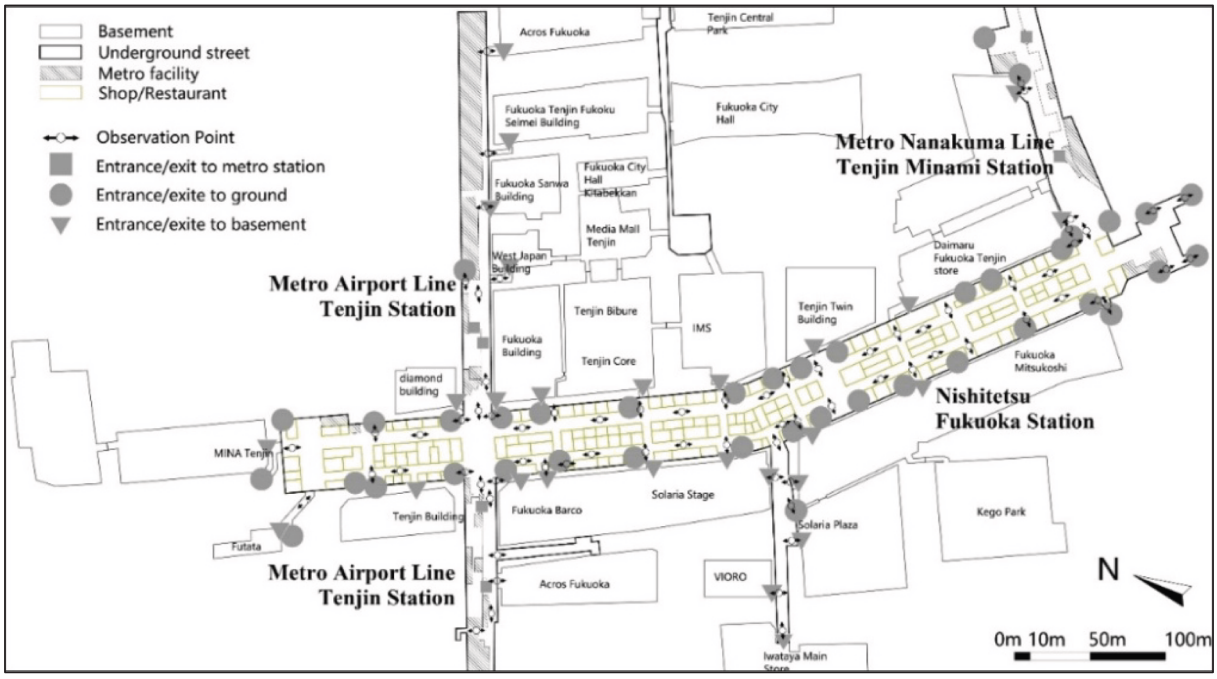
To account for the influence of cloud cover and other meteorological conditions on solar exposure and pedestrian decision-making, pedestrian route choice was classified into two behavioral strategies: a shortest-distance strategy and a utility-based strategy.

During periods of significant cloud cover, which mitigates solar radiation, pedestrians were assumed to adopt a distance-priority strategy, selecting the route with the shortest travel time represented by the red path in Fig. 7. Conversely, under clear-sky conditions with direct sunlight, pedestrians are expected to adopt a utility-based strategy, choosing paths based on a trade-off between sun exposure and physical cost (green path in Fig. 7).

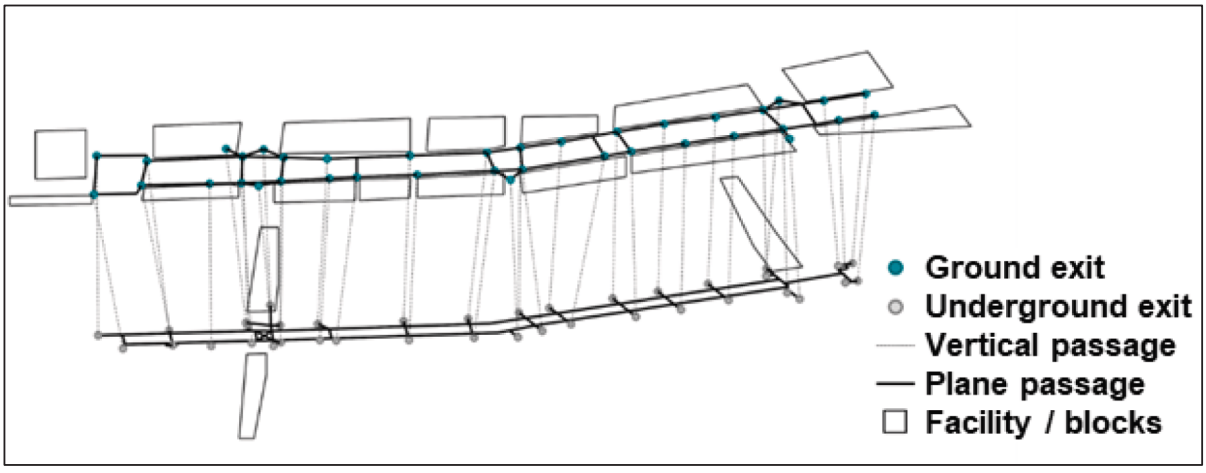
For each OD pair between origin i and destination j, both shortest-distance and utility-based routes were generated using the Dijkstra algorithm under different solar exposure conditions. The proportion of pedestrian traffic along a ground-level street segment l at time t was determined as follows:

$$P_{ijlt} = Y_t L_{ijlt} + (1\text{-}Y_t)\, K_{ijlt} \tag{2}$$

where P_{ijlt} denotes the proportion of trips from origin i to destination j



(a)



(b)

Fig. 3. Three-dimensional pedestrian network of the study area: (a) underground shopping street layout; (b) connections between the ground-level and underground pedestrian networks.

traversing segment l at **time** t; Y_t is the normalized proportion of sunlight duration (range: 0–1) at time t; L_{ijlt} and K_{ijlt} denote the link usage frequencies under utility-priority and distance-priority decision-making.

By aggregating the sun-exposure durations associated with both route-choice strategies, the expected sun-exposure duration for each OD pair was calculated as

$$E_{ijt} = Y_t \sum_l P_{ijlt} D_{lt} \tag{3}$$

The expected sun-exposure duration during heatwave periods was calculated as

$$W_{ijt} = Y_t B_t \sum_l P_{ijlt} D_{lt} \tag{4}$$

Accordingly, the expected duration of pedestrian exposure to dangerous heat conditions was obtained as

$$R_{ijt} = \begin{cases} W_{ijt}, & \text{if } \mathrm{WBGT}_{\mathrm{shade}} < h_{\mathrm{danger}} \le \mathrm{WBGT}_{\mathrm{sunlight}} \\ \sum_l A_l P_{ijlt}, & \text{if} h_{\mathrm{danger}} \le \mathrm{WBGT}_{\mathrm{shade}} \\ 0, & \text{if } h_{\mathrm{danger}} > \mathrm{WBGT}_{\mathrm{sunlight}} \end{cases} \tag{5}$$

where B_t is a Boolean variable indicating whether the WBGT under sunlit outdoor conditions exceeds the danger threshold. Furthermore, h_{danger} is the WBGT threshold corresponding to the danger level defined by the JME warning levels, and A_l represents the total walking and waiting time at link l. If the shaded WBGT exceeds the warning threshold, the entire walking and waiting time on link l was regarded as exposure to dangerous heat conditions.

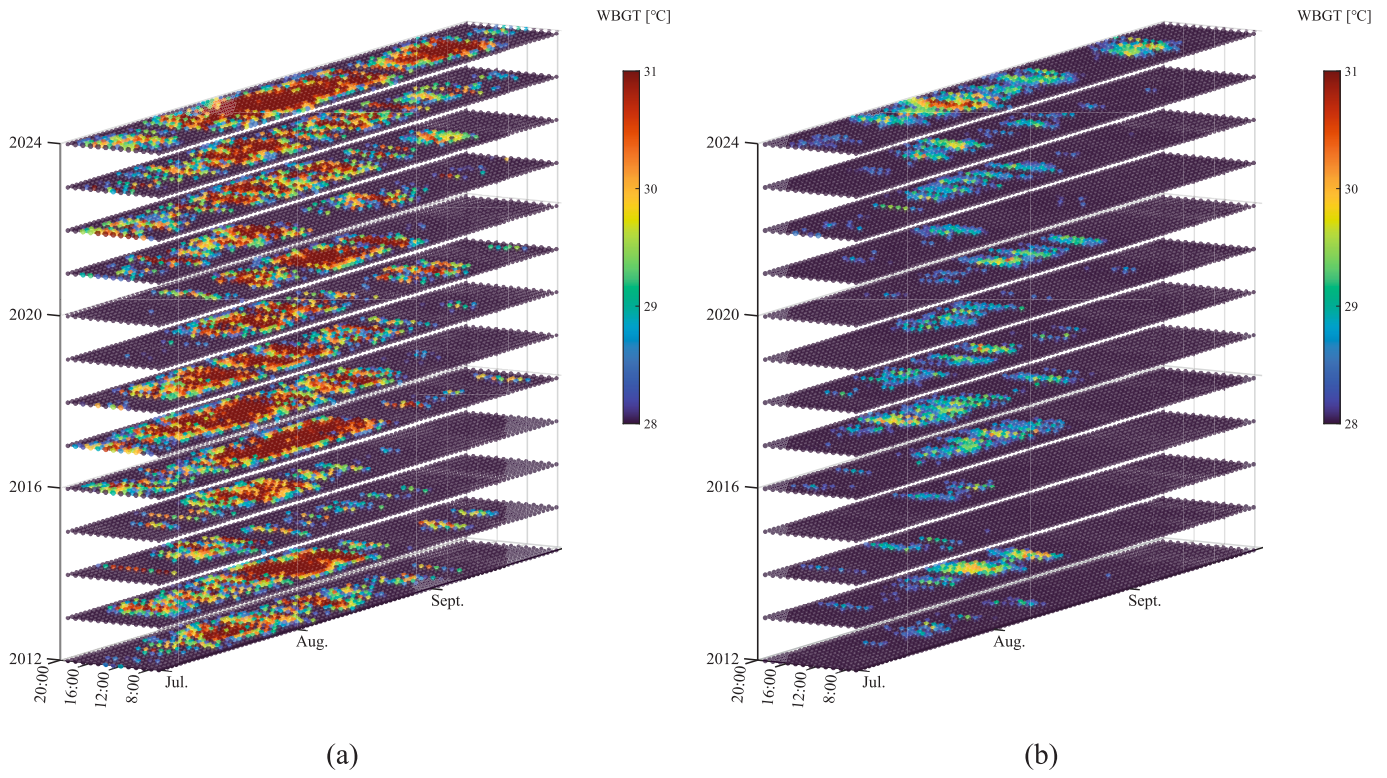


Fig. 4. Temporal variation in WBGT during the simulation period: (a) sunlit conditions; (b) shaded conditions.

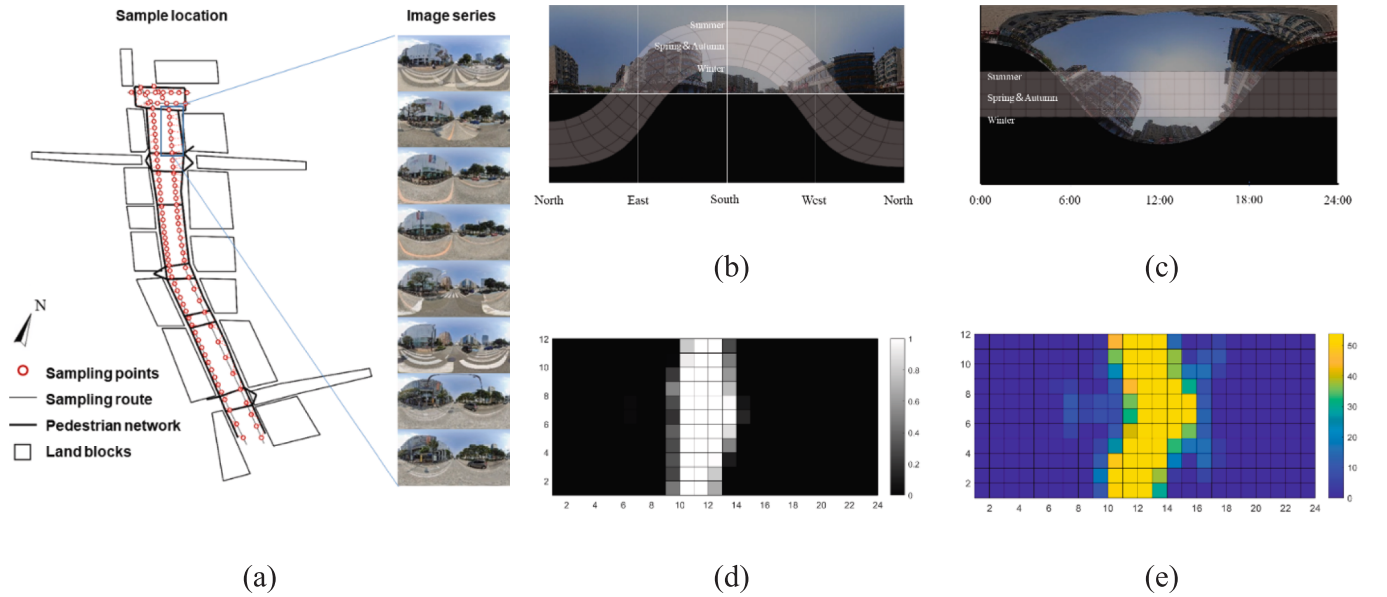


Fig. 5. Method of estimating sun-exposed areas' spatiotemporal distribution in ground-level pedestrian networks: (a) streetview image sampling; (b) overlap solar trajectory; (c) projection transformation; (d) sun-exposed probability matrix; (e) sun-exposed length f street link.

2.4. Indicators for evaluating the UPS's contribution to heat-risk mitigation

Two types of indicators are defined to quantitatively assess UPS's contribution to mitigating users' potential sun-exposure risk during summer heatwaves.

2.4.1. System-level indicators

Sun-exposure risk mitigation index (SRMI): The cumulative exposure reduction achieved by the UPS at time t is calculated as the sum of all pedestrians' expected exposure duration:

$$\mathrm{SRMI}_t = \sum_i \sum_j T_{ijt} W_{ijt} \tag{6}$$

Sunshine mitigation index (SMI):is defined as the cumulative walking and waiting time of UPS users under sunlit outdoor conditions.

$$\mathrm{SMI}_t = \sum_i \sum_j T_{ijt} E_{ijt} \tag{7}$$

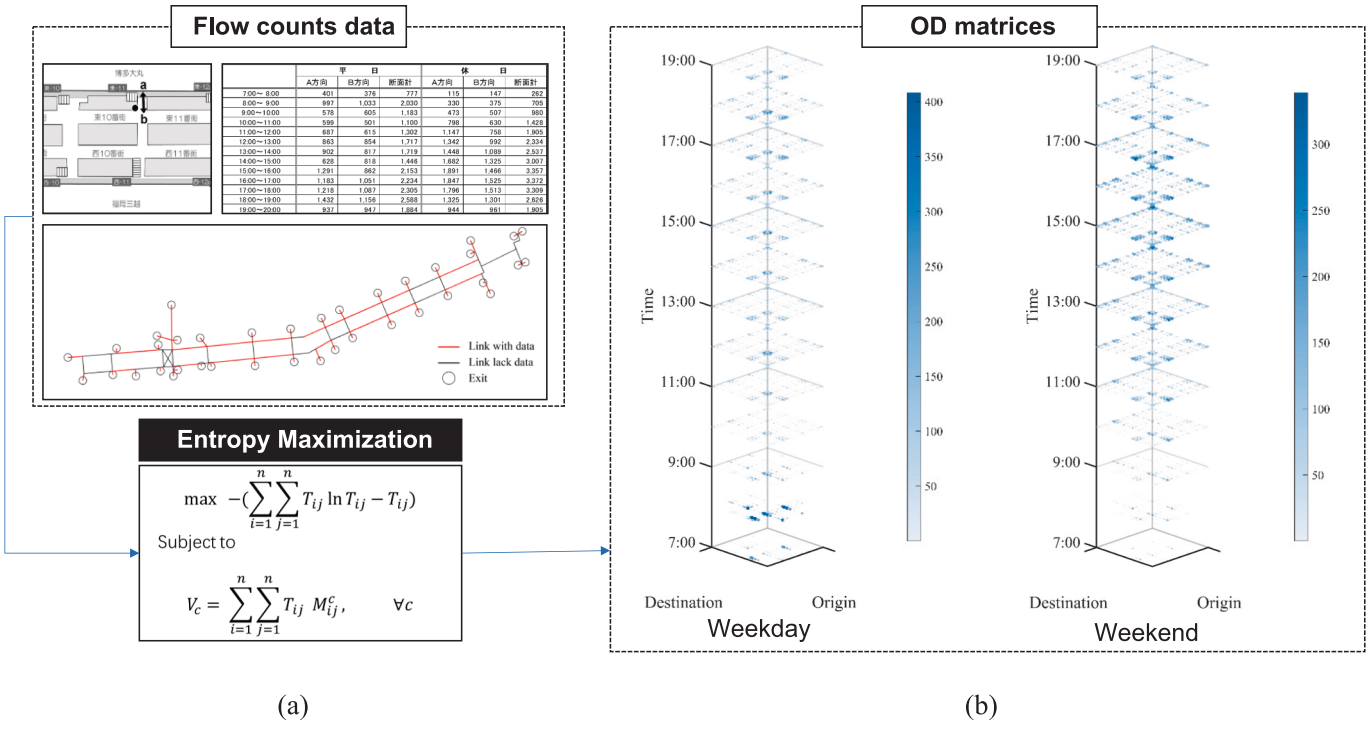


Fig. 6. Workflow for estimating hour-specific OD matrices in the underground pedestrian network: (a) entropy maximization method; (b) estimated hour-specific OD matrices for weekdays and weekends.

Table 1
Equivalent distance conversion coefficients for pedestrian facility and environmental factors.

Parameters	Unit	Coefficients (α_n)	Reference
Length (d_l)	Meter	1	Hiroyuki et al., 2002
Vertical movement (stairs) (S_l)	Step	1.2(upwards) / 0.6(down)	
Traffic signal (W_l)	Signal	30	Nishimura and Osaragi, 2009
Sun-exposed duration (D_{lt})	Second	0.196 (building shadow)/ 0.098 (tree shadow)	Melnikov et al., 2022

Extreme-heat-exposure mitigation index (EMI) is defined by pedestrians' accumulative exposure duration in the outdoor environment of danger-level:

$$\mathrm{EMI}_t = \sum_i \sum_j T_{ijt} R_{ijt} \tag{8}$$

These indicators represent the overall effectiveness of the UPS in responding to climatic impacts and can be used to analyse temporal variations across multiple timescales.

2.4.2. *Spatiotemporal indicators*

Risk exposure frequency (REF): is used to characterize the spatiotemporal distribution of UPS's contribution in reducing sun-exposed risk in the ground-level pedestrian network. The REF is defined as the cumulative additional sun-exposure duration on street link l in the absence of the UPS:

$$\mathrm{REF}_{lt} = Y_t \sum_i \sum_j T_{ijt} P_{ijlt} D_{lt} \tag{9}$$

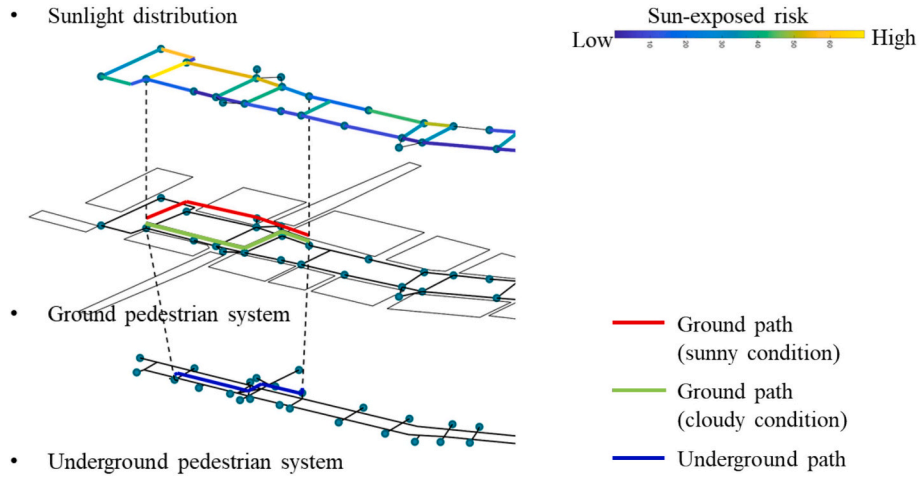


Fig. 7. Pedestrian path simulation based on sun-exposed risk area and cloudy conditions.

This indicator is a link-specific value reflecting the time-varying relationship between pedestrian paths and heat risk associated with sunlight exposure.

Risk mitigation gain (RMG): is defined by the cumulative risk avoidance of pedestrians between a specific OD pair:

$$\mathrm{RMG}_{ijt} = T_{ijt} W_{ijt} \tag{10}$$

This indicator generates a matrix used to illustrate the UPS's contribution to different travel groups. RMG values between UPS entrances and exits can be transferred to represent the travel group gains between the urban blocks and major facilities by:

$$\mathrm{RMG}_{qpt} = \left(G_{ijt} I\right)^T I \tag{11}$$

where q and p represent the indices of urban block. I is an n × m transition probability matrix representing the probabilities between the UPS entrances/exits and the urban blocks. T denotes matrix transposition.

Network support efficiency (NSE): Using underground network components (nodes, links, and entrances/exits) as spatial units, this indicator evaluates the spatial distribution of UPS support efficiency by allocating avoided exposure risk to each network component:

$$\mathrm{NSE}_{kt} = \sum_i \sum_j G_{ijt} U_{ijkt} \tag{12}$$

where U_{ijkt} represents the frequency with which trips between OD pair (i, j) traverse underground network element k (Gu & Osaragi, 2017). In this study, UPS entrances/exits were selected as the target network elements to identify the vertical access points contributing most substantially to heat-risk mitigation.

3. Result and analysis

3.1. Global sunlit exposure in different years

Fig. 8 illustrates the cumulative reduction in pedestrian exposure to direct sunlight and to the sunlit streets with heat risk achieved by the UPS in the study area from 2012 to 2024. Each cell in the figure represents the results of a one-hour simulation within the study period. The horizontal position of a cell corresponds to the simulated date, while its vertical position indicates the time of day. The year index is labeled at the corresponding noon position (12:00). In the figure, cells in warm colors denote longer cumulative exposure durations, whereas cells in cool colors represent shorter durations.

Fig. 8(a) clearly illustrates the patterns of the system's sun-exposure mitigation (SMI) contribution across different time scales. The daily variation in sunlight exposure duration shows a periodic fluctuation that peaks around noon. Along the horizontal axis, periods with high cumulative exposure duration expand in a weekly cycle, reflecting the system's differential response to travel demands on weekdays versus weekends. Furthermore, intermittent gaps along the horizontal axis indicate that the system periodically provided little or no mitigation to sunlight exposure owing to weather influences such as rainfall or cloud cover. Interannual variations in SMI are evident. For instance, system effectiveness during the 2020–2022 period was generally lower compared to other years. This demonstrates how the system's climatic sheltering function responds over longer time scales under the combined influence of climate variability and public health emergencies.

Fig. 8(b) presents the simulated results of extreme-heat-exposure mitigation index (EMI) in the outdoor environment during heatwave periods. Compared with the general sunlight scenarios, the daily range of non-zero EMI is shorter. However, the period of high cumulative exposure around noon remains consistent. This is related to lower temperatures in the morning and evening, when walking under direct sunlight does not induce dangerous heat stress ($\mathrm{WBGT}_{\mathrm{sunlit}} < 31$).

In a horizontal direction, the non-response periods of system are longer. This indicates that cooling effects from rainfall or typhoon may persistently interrupt the system's response to heatwave periods. The cyclical pattern of high cumulative exposure duration related to travel demand remains observable, but only in years with prolonged heatwaves, such as 2024. Interannual variation is particularly pronounced. The number of days with a high EMI fluctuated widely, ranging from 5 days in 2014 to 60 days in 2024. Similar to the pattern in Fig. 8(a), overall system effectiveness was lower between 2020 and 2022. However, the date distribution of high EMI during these years did not differ significantly from that observed in other years.

The results of SRMI are similar to those of the EMI, while the range of the SRMI is smaller (Fig. 8(c)). This is because EMI also includes the scenarios in which the WBGT on the shaded streets exceeds the danger-level threshold. To further distinguish these periods, Fig. 8(d) visualizes the ratio of EMI to SRMI for each time interval. White cells indicate "none", representing periods in which the study area with no direct sunlight due to rainfall or cloud-overcast conditions. Orange and blue cells denote ratios of 1 and 0, respectively. These cells imply whether sun-exposed road segments posed a heat risk during the corresponding period. The figure further shows that alternating blue and white regions along the date axis indicate frequent summer rainfall and cloudy weather. The associated cooling effect makes the walking on sunlit streets does not pose a high thermal risk, which leads to interruptions in the system's response to sun-exposed heat risk. This phenomenon is particularly obvious in 2014. In years with less frequent rainfall and fewer cloud-overcast conditions, such as 2016 and 2024, the heat risk associated with walking in sunlight persisted for nearly one month. In these years, the UPS was more effective in mitigating the impact of heatwaves on the pedestrian system. The temporal distribution of this ratio provides a clear understanding of the cyclical pattern of the UPS's response to climatic stress. A ratio greater than 1 indicates that even shaded road segments reached the thermal danger level. Under such conditions, a UPS without indoor air-conditioning system cannot reduce pedestrians' heat exposure risk merely by providing shaded paths. The system completely loses its heat-risk mitigation function when the climatic stress exceeds this threshold.

Purple cells in Fig. 8(d) show that the ratio exceeded 1 on multiple occasions in early August 2024. UPSs without air-conditioning may face a risk of functional failure under future climate change. Consequently, the construction and upgrading of indoor air-conditioning systems in these UPSs will become an important issue that managers and planners cannot afford to overlook.

3.2. Spatiotemporal distribution of UPS's climate resilience response

Based on the indicators defined in Section 2.4.2, Fig. 9 quantitatively visualizes the spatiotemporal distribution of the UPS's contribution to mitigating sunlight exposure during heatwave periods. The vertical stacking of the layers allows an examination of the annual variations in Risk Exposure Frequency (REF), Risk Mitigation Gain (RMG), and Network Support Efficiency (NSE)..

In Fig. 9(a), the color of each street segment indicates the spatial distribution of sunlight-exposure area under heatwave conditions. Brighter segments represent paths where pedestrians were more frequently exposed to heat risk caused by direct sunlight. This result indicates the vulnerability of the street network to heat stress. It can be observed that the year-to-year changes in vulnerability are not synchronous across different street segments. The vulnerability of paths with frequent heat exposure shows more significant variation over time.

Fig. 9(b) illustrates the contribution of the UPS to different travel groups between street blocks based on the RMG defined in the previous section. The color of each connecting line represents the risk mitigation gain received by each travel group using the UPS, while the line width indicates the number of travelers. To improve the visualization, connections with benefits less than 1% of the maximum value have been omitted. By examining the length, width, and color of these connections, the relationships among travel distance, user volume, and risk

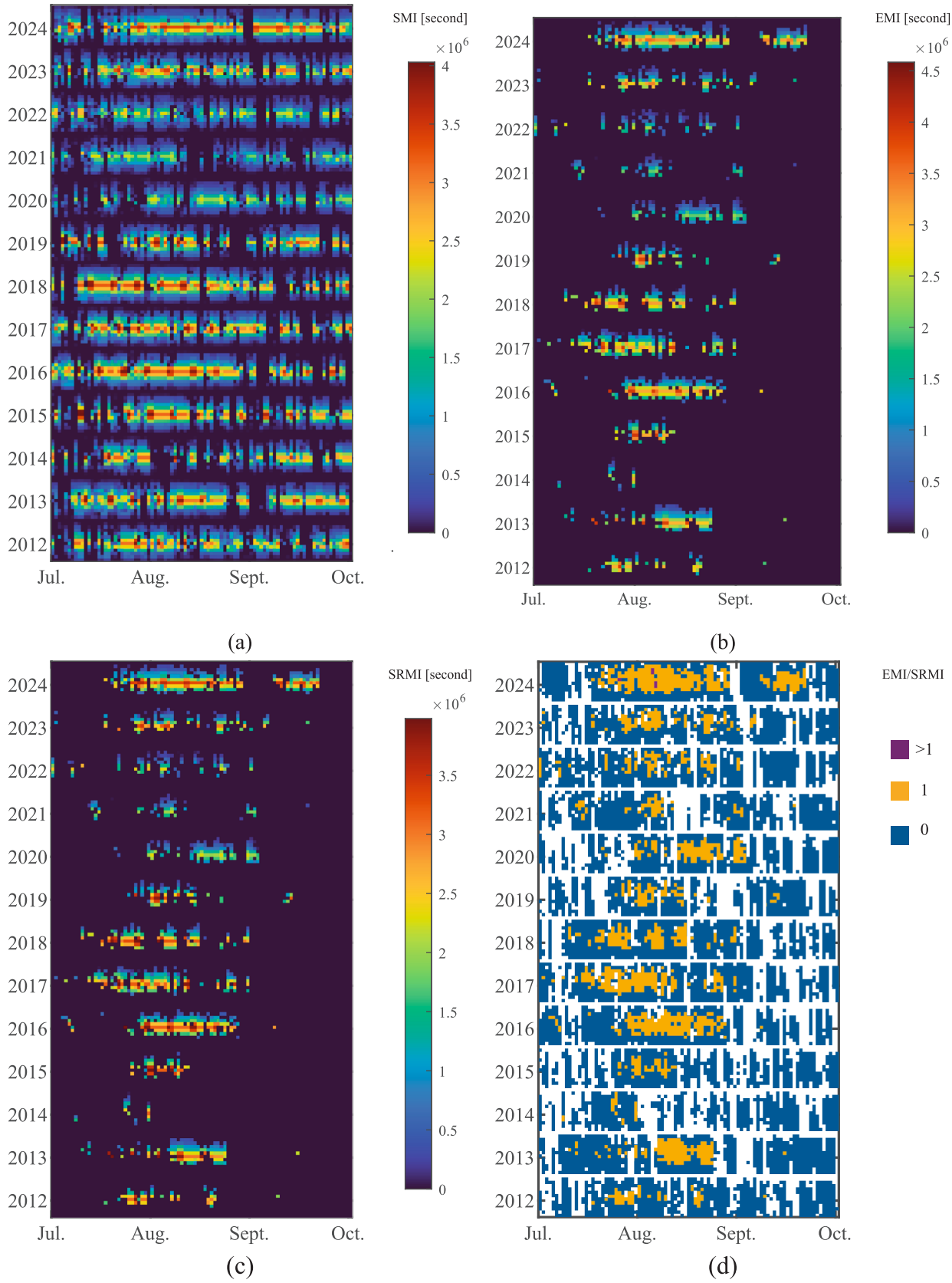


Fig. 8. Temporal distribution of system-level indicators: (a) sunlight-exposed mitigation index (SMI); (b) extreme-heat-exposure mitigation index (EMI); (c) sunlight-exposure risk mitigation index (SRMI);(d) ratio of EMI to SRMI.

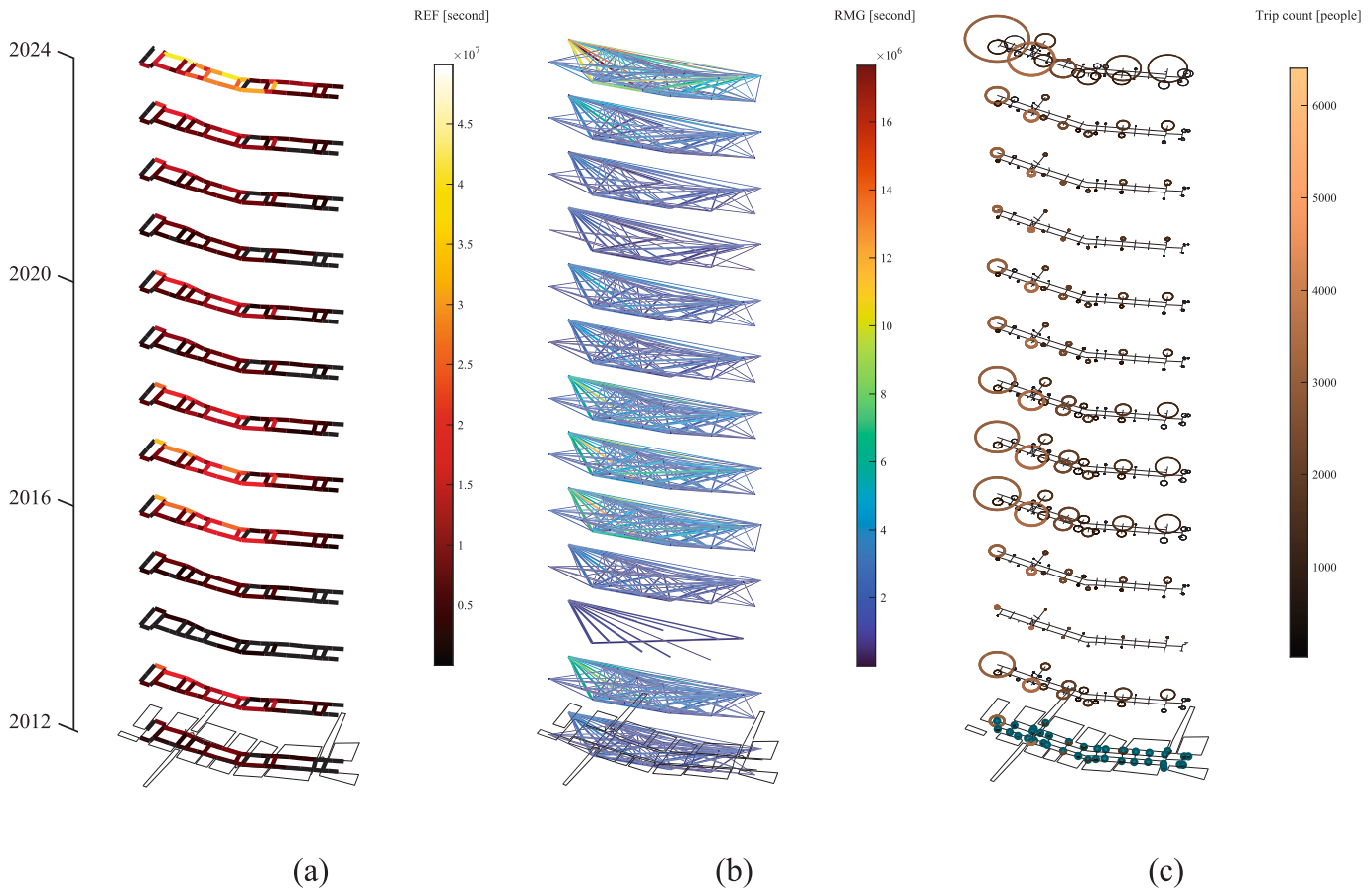


Fig. 9. Spatiotemporal distribution of UPS climate resilience indicators: (a) risk exposure frequency (REF); (b) risk mitigation gain (RMG); (c) network support efficiency (NSE).

mitigation gain can be identified. It was found that pedestrian volume and travel distance are not the sole determinants of RMG. The interannual variation in the spatial distribution of RMG also shows spatial inconsistency. This pattern reflects the inequity in the system's service provision.

Fig. 9(c) represents the support efficiency in the UPS climate resilience response. Support efficiency is quantified by the cumulative sun-exposure risk duration of paths passing through specific entrances/exits. The radius of each circle indicates the efficiency of the entrance/exit at its center, while the color and line width of the circle represent the volume of pedestrians passing through it. The results show that the spatial distribution of component support efficiency is relatively consistent with the risk exposure pattern shown in Fig. 9(a). Furthermore, entrances/exits with higher support efficiency often serve as the primary entrances for high-RMG user groups entering the UPS. Consequently, the year-to-year variation in entrance/exit efficiency also exhibits spatial heterogeneity, with high-efficiency entrances/exits showing greater interannual variability. This finding helps identify critical nodes within the underground spaces for enhancing the climate resilience performance of the UPS.

3.3. *Environmental factors affecting the temporal variation in SRMI*

This section examines how periodic variations in environmental factors influence the temporal variation in the UPS resilience response, using a time-series statistical analysis of SRMI.

3.3.1. *Temporal distribution of SRMI*

Simulation results under different temporal scenarios were aggregated by summing SRMI across three temporal dimensions (year, date, and hour) and then transformed into a two-dimensional matrix. To minimize the influence of travel demand differences between weekdays and weekends, the date dimension was aggregated using a seven-day interval. Fig. 10 presents the temporal distribution surfaces of SRMI for each pairwise combination of the three dimensions. The height and color of the surface represent the cumulative SRMI of scenarios with the same time labels.

As shown in Fig. 10(a), the SRMI distribution across the “date–year” dimensions exhibits a clear multi-peak pattern before 2019. This pattern is associated with the uncertainty in the interannual variability in rainfall occurrence. Influenced by factors such as air temperature and daylight duration, higher SRMI values are mainly concentrated between mid-July and late August. In recent years, with the acceleration of climate change, these multiple peaks have gradually merged into a single extended peak centered in early August and lasting for approximately one month. In addition, SRMI levels in September show an increasing trend in recent years.

Fig. 10(b) shows a two-dimensional matrix, with rows representing the time of day and columns representing the date. The data exhibit an approximately Gaussian distribution along both dimensions: peaking in early August for the date dimension and at noon (12:00) for the time dimension. When visualized as a spatial surface, the data forms a three-dimensional Gaussian surface. The surface displays a distinct peak centered at the intersection of early August and noon, corresponding to the maximum data values. In recent years, an emerging peak in mid-September appears as a protrusion on the surface. This pattern may become more pronounced in the future as climate conditions continue to change.

Fig. 10(c) shows the temporal distribution, with rows representing years and columns representing the time of day (in hours). Along the

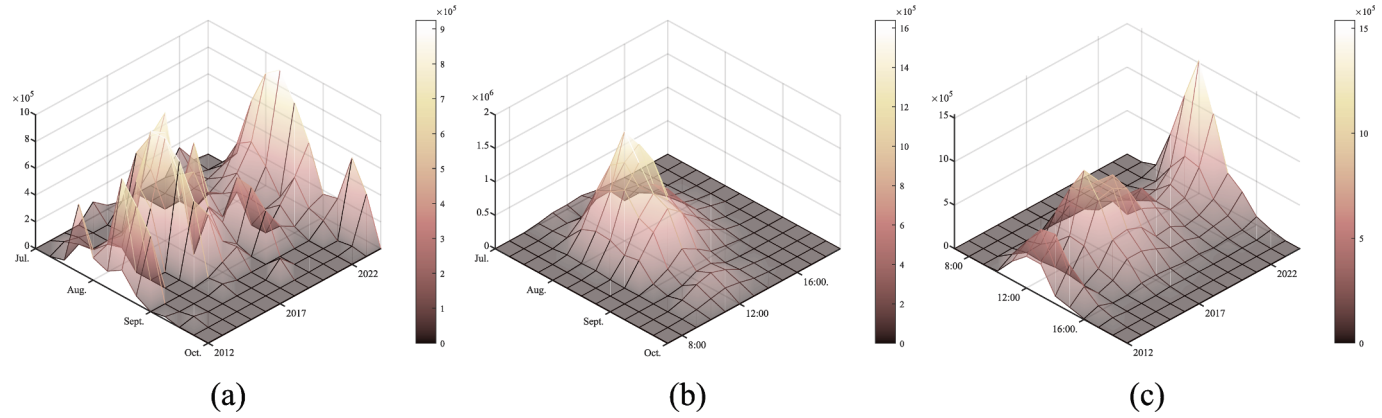


Fig. 10. Temporal variation of SRMI across different temporal scales: (a) year–date; (b) date–hour; and (c) hour–year.

hour dimension, the data exhibit a Gaussian distribution, peaking at around noon (12:00), mirroring the pattern observed in Fig. 10(b). However, a distinct divergence emerges along the year dimension. Due to notably suppressed data values in the years 2014 and 2021, the overall spatial surface reveals a characteristic saddle-shape. A Gaussian distribution persists along the time axis, with peaks consistently occurring around noon. Conversely, along the year axis, the surface exhibits a bimodal depression, creating two saddle points at the noon of 2014 and 2021.

The above analysis indicates that differences in UPS's resilience performance across time dimensions increase the complexity and uncertainty of its temporal variation. As a result, risk exposure simulation based on a single time scenario is insufficient to accurately capture the long-term climate resilience performance of the UPS.

3.3.2. *Seasonal climate factors affecting UPS resilience performance*

This section analyzes how variations in climatic factors, specifically, air temperature and hourly sunshine duration, across different time dimensions affect SRMI. Air temperature is a key factor related to heatwave scenarios, while sunshine duration influences the duration of pedestrian sunshine exposure. Fig. 11 shows the cumulative changes in SRMI across the year, date (week), and hour dimensions. By overlaying these results with the mean air temperature and the average sunshine duration within each time unit, the correlation between SRMI and climatic factors is examined.

The bar charts in Fig. 11(a) and (b) illustrate the interannual variation in the cumulative SRMI. As mentioned before, two obvious declines occur in 2014 and 2021. Comparison with the interannual trends in mean air temperature and daily cumulative sunshine duration shows a good agreement between these climatic factors and SRMI. In some periods, such as 2014–2015 and 2019–2020, changes in cumulative sunshine duration more closely match the variation in SRMI.

Fig. 11(c) and (d) present the seasonal variation of daily mean SRMI using a weekly time unit. SRMI follows an approximately normal distribution, with a peak in early August. This pattern is related to the seasonal trends of air temperature and daily cumulative sunshine duration. On dates when the mean air temperature is below 30 °C or the cumulative sunshine duration is shorter than 6 h, the daily mean SRMI shows a clear decline. In addition, the rebound of SRMI observed in mid-September aligns more closely with fluctuations in cumulative sunshine duration. This correspondence suggests that the recent extension of UPS's performance into mid-September may be associated with reduced rainy conditions and increased sunshine duration.

Fig. 11(e) and (f) show the hourly variation of SRMI and its relationship with air temperature and sunshine duration. As noted earlier, SRMI exhibits a single-peak pattern with a maximum around 12:00. However, the distribution is asymmetric between morning and afternoon. The increase in SRMI during the morning is steeper than the decline observed in the afternoon. This pattern more closely follows the temporal change in air temperature. Sunshine duration reaches relatively high levels from around 09:00, but this increase does not lead to a noticeable rise in SRMI.

Overall, the results indicate that the influence of climatic factors on SRMI reflects the combined effects of multiple variables. Air temperature and sunshine duration show different degrees of association with SRMI across different time scales.

To examine the influence of climatic factors on SRMI, separate univariate linear regression analyses were conducted using hourly observations as independent samples. As shown in Table 2, temperature (adjusted $R^2 = 0.1401$) and sunshine duration (adjusted $R^2 = 0.0839$) exhibited relatively low explanatory power for variations in SRMI based on hourly observations. These results suggest that climatic variables alone are insufficient to fully explain the heat-mitigation performance of the UPS. One possible explanation is that temporal fluctuations in UPS usage introduce substantial uncertainty, thereby reducing the sensitivity of SRMI to changes in temperature and sunshine duration.

To further investigate this relationship, annual, weekly, and hourly mean values of temperature, sunshine duration, and SRMI were calculated for each temporal scale and analysed separately. At the annual scale, the explanatory power of temperature and sunshine duration increased markedly, with adjusted R^2 values of 0.755 and 0.475, respectively ($p < 0.001$ and $p = 0.007$). This finding indicates that the heat-exposure mitigation effect of the UPS is more responsive to long-term temperature variations associated with climate change, whereas its response to fluctuations in sunshine duration is comparatively weaker.

The strongest relationships were observed at the weekly scale, where temperature and sunshine duration explained 78.9% and 79.3% of the variance in SRMI, respectively. This suggests that UPS resilience performance is most sensitive to seasonal climatic variations. A likely reason is that weekly averaging reduces the influence of short-term fluctuations in pedestrian volumes and interannual disturbances, such as mobility suppression during the COVID-19 pandemic, thereby revealing the underlying climatic effects more clearly.

By contrast, the explanatory power of hourly mean climatic variables was substantially lower (adjusted $R^2 = 0.511$ for temperature and 0.251 for sunshine duration). This result implies that the temporal mismatch between outdoor climatic conditions and the distribution of UPS users significantly weakens the responsiveness of SRMI to environmental changes. This interpretation will be further examined in the subsequent factor analysis. Overall, the results highlight a pronounced temporal-scale dependence in the climate–SRMI relationship, indicating that the resilience performance of the UPS is governed not only by climatic forcing but also by the temporal dynamics of pedestrian demand.

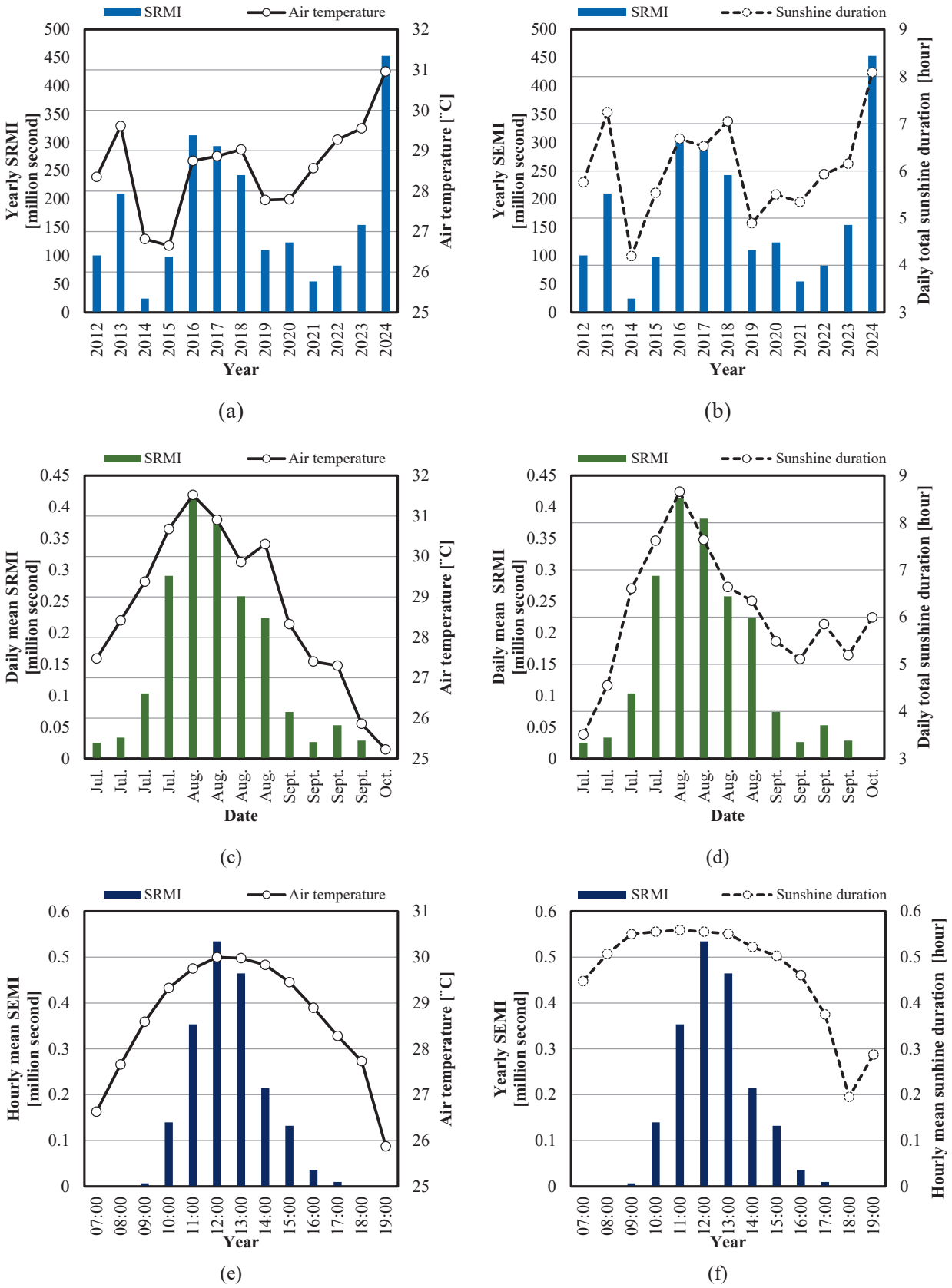


Fig. 11. Temporal correspondence between SRMI and climatic factors across different temporal scales: (a-b) years; (c-d) weeks; and (e-f) hours.

Table 2
Results of the univariate linear regression analyses between SRMI and climatic variables.

Predictor variable	Adjusted R^2	P value	N
Hourly observed air temperature	0.1401	<0.001	15,548
Hourly observed sunshine duration	0.0839	<0.001	15,548
Annual mean air temperature	0.756	<0.001	13
Annual mean sunshine duration	0.457	0.007	13
Weekly mean air temperature	0.789	<0.001	12
Weekly mean sunshine duration	0.793	<0.001	12
Hourly mean air temperature	0.511	0.004	13
Hourly mean sunshine duration	0.251	0.46	13

3.3.3. *Effects of urban morphological factors on SRMI*

The length of shaded routes generated by buildings and trees is an urban morphology-related variable that influences the temporal variation of SRMI. Changes in solar position cause the sunlit path ratio to vary across both date and hour dimensions. Here, the sunlit path ratio refers to the proportion of route length receiving direct solar radiation relative to the total street segment length under clear, cloudless conditions.

Fig. 12(a) shows that the sunlit path ratio in ground networks gradually decreases from early July to late September. However, this trend does not exhibit a clear correspondence with changes in SRMI. Fig. 12(b) presents the temporal relationship between these two variables across the hourly dimension. Both variables exhibit approximately normal temporal distributions. A key difference is that the peak of the non-shaded route ratio occurs earlier, at around 11:00, than the SRMI peak. This pattern differs from the relationship observed between air temperature and SRMI in Fig. 11(c). The seasonal and hourly variations in the sunlit route ratio exhibited negligible explanatory power for SRMI, with adjusted R^2 values of 0.001 and 0.016, respectively ($p = 0.914$ and $p = 0.699$). This suggests that changes in shadow distribution resulting from urban morphology are not a primary driver of UPS resilience performance within the study area. Compared with the climatic factors, the influence of ground-level shading patterns on the heat-exposure mitigation effect of the UPS appears to be relatively limited. In addition, time periods with similar sunlit path ratios can correspond to markedly different SRMI values. In Fig. 12(b), two pairs of time periods with comparable sunlit ratios are highlighted in orange and green. During the afternoon periods (13:00 for the green pair and 15:00 for the orange pair), the hourly SRMI is 3.3 times and 19.8 times higher than in their corresponding morning periods of 10:00 and 09:00.

To further examine the resources of this disparity, the spatial distribution of risk exposure frequencies (REF) in the corresponding period pairs were analyzed in more detail. Fig. 13 visualizes the spatial distribution of REF using color intensity, where warmer colors indicate higher REF. The width of each street segment represents the length of sun-exposed routes. In addition, the network support efficiency of each exit is used to indicate where trips are more likely to be exposed to solar-related risk. Although the sunlit path ratio is the same (20%) at 09:00 and 15:00, the spatial pattern differs markedly. Areas with higher sunlit ratios, represented by wider segments, shift from a central concentration to the upper part of the study area. At the same time, risk-exposed trips become more concentrated at the upper exits (A1-B1), while cumulative exposure risk associated with other exits (A2-B2 and A3-B3) decreases substantially. As a result, the upper part of the street network exhibit higher exposure frequency. The spatial overlap between solar exposure scenarios and travel demand leads to a stronger resilience contribution during the 15:00 period.

Under conditions where the sunlit path ratios reach 40% (Fig. 13(c) and 13(d)), further differences are observed. At 10:00, the sunlit route proportion in the upper area is lower (narrower links), allowing pedestrians traveling between the two main trip origins (C1 and C2) to remain in shaded areas for longer distances. In contrast, at 15:00, sun-exposed areas shift from the central to the upper region. This increases the duration of sun exposure along routes connecting the major trip origin (D1) with the remaining destinations, leading to a higher overall SRMI. These results indicate that the sunlit path ratio alone does not determine the temporal distribution of SRMI. The spatial distribution of pedestrian flows and the spatial mismatch between trip demand patterns and sun-exposed risk distribution must also be considered.

Fig. 14 further illustrates how the spatial–temporal mismatch between pedestrian paths and heat-exposure risk distributions influences the responsiveness of the UPS. In Fig. 14(a) and (b), the blue and yellow histograms represent the frequency of pedestrian movements across links with different sunlit route-length intervals at 9:00 and 15:00, respectively. The corresponding dashed and solid lines indicate the cumulative SMI and SRMI associated with each sunlit route-length interval. The results show that pedestrian movements at 15:00 are more concentrated on links with relatively long sunlit route lengths, particularly within the 30–40 m intervals. Consequently, both cumulative SRMI (full curve in Fig. 14(a)) and cumulative SMI (dash curve in Fig. 14(b)) within these intervals are substantially higher than those observed at 9:00. However, compared with the distribution of cumulative SMI, the cumulative SRMI at 9:00 remain consistently lower across all risk intervals. This difference can be attributed to the relatively lower air

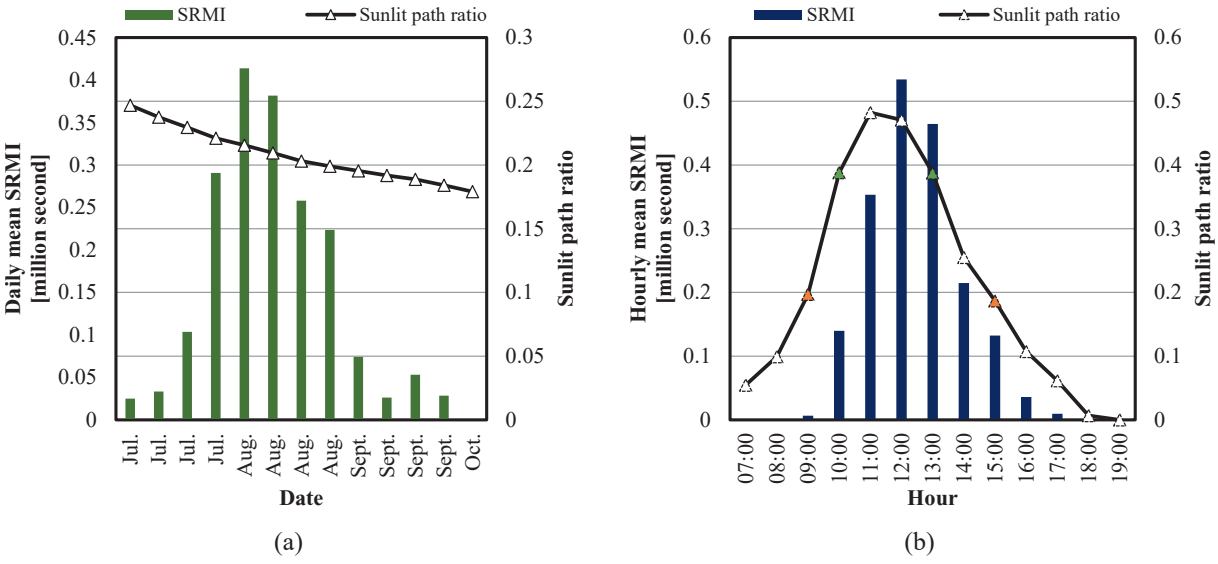


Fig. 12. Temporal relationships between SRMI and urban morphology factors across: (a) the date dimension; (b) the hourly dimension.

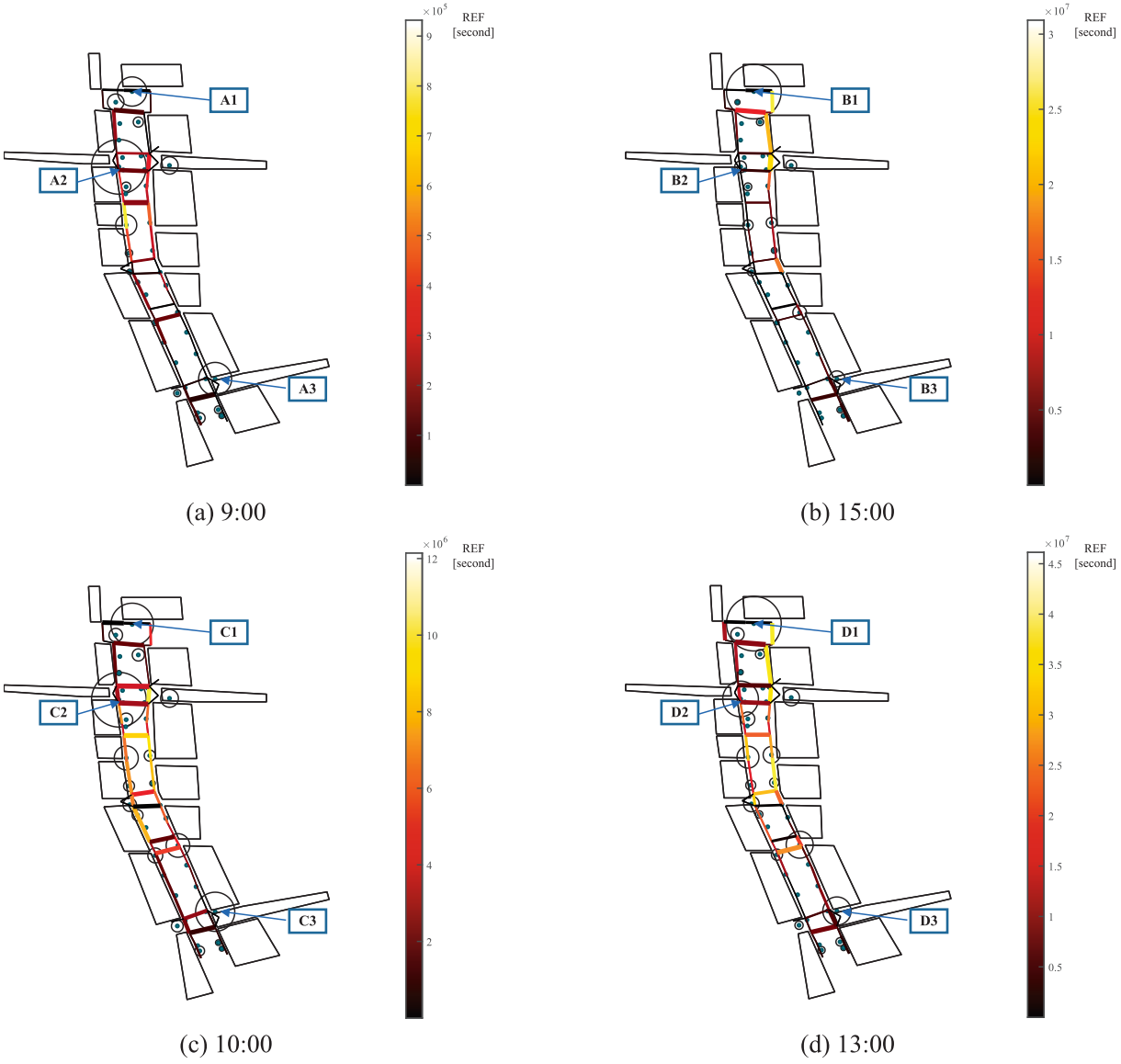


Fig. 13. Spatial distribution of risk exposure frequency (REF) during four representative periods: (a) 9:00; (b) 15:00; (c) 10:00; (d) 13:00.

temperatures during the morning period, when most sunlit links do not exceed the heat-index threshold corresponding to the danger-level. This interpretation is consistent with the temperature patterns presented in Fig. 11(e).

Fig. 14(c) and (d) provide a comparison between the 10:00 and 13:00 periods. In both cases, the frequency distributions exhibit pronounced peaks within the 35–50 m intervals, corresponding to links with relatively high exposure risk. This pattern arises because the simulation framework converts the expected waiting time at signalized crossings into an equivalent travel distance under sun exposure. During these periods, pedestrian crossing links are less affected by building shadows (Fig. 13(c) and (d)), while increasing demand for cross-street movements along major roads results in a larger proportion of pedestrian trips occurring on highly sun-exposed links. As a result, both SRMI (full curve in Fig. 14(c)) and SMI (dash curve in Fig. 14(d)) display bimodal distributions across sunlit route-length intervals. Notably, the differences in cumulative SRMI between 10:00 and 13:00 are more pronounced than those observed for cumulative SMI. This finding is also consistent with Fig. 11(e), which shows higher average air temperatures at 13:00 than at 10:00. Taken together, these results indicate that the effectiveness of the UPS in mitigating sun-exposure risk during heatwave conditions is jointly determined by the spatiotemporal coupling between pedestrian movement patterns and exposure-risk distributions, as well as by air temperature.

By establishing a spatially explicit resilience-performance allocation mode, the proposed framework enables researchers to identify how the heterogeneous distribution of driving factors across spatial units dynamically shapes system resilience. This capability provides a deeper understanding of the mechanisms through which spatial heterogeneity affects the climate-adaptive performance of pedestrian infrastructure.

3.4. Behavioural factors affecting the spatial and temporal distribution of SRMI

This section uses the spatial and temporal analyses to examine how behavioural factors (such as UPS usage volume and trip distribution) affect the spatiotemporal dynamics of UPS response performance.

3.4.1. Relation with passenger usage

Fig. 15(a) shows the relationship between annual SRMI and the passenger volume of the three metro/railway stations. The results first suggest that the decline observed in 2014–2015 is not driven by changes

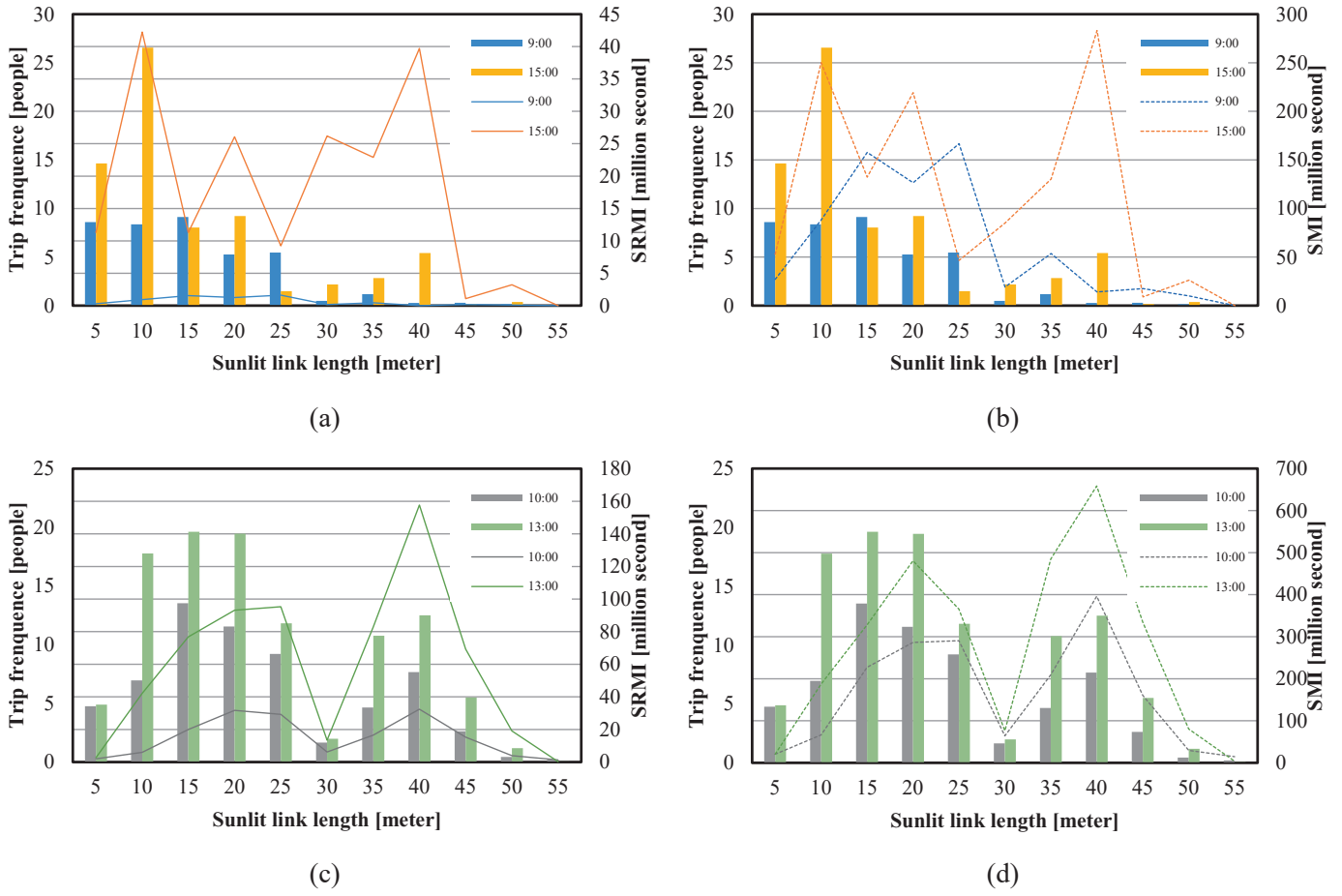


Fig. 14. Frequency distributions of pedestrian trips, cumulative SMI, and cumulative SRMI across different sunlit route-length intervals during representative periods: (a) SRMI at 09:00 and 15:00; (b) SMI at 09:00 and 15:00; (c) SRMI at 10:00 and 13:00; (d) SMI at 10:00 and 13:00.

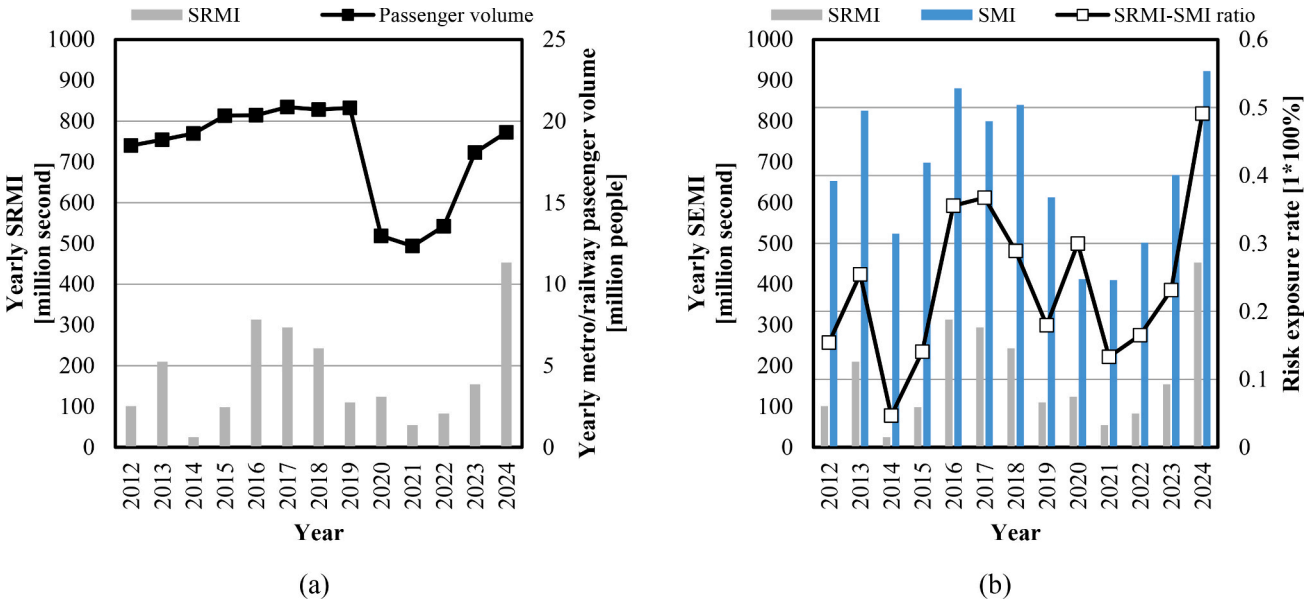


Fig. 15. Relationship between annual SRMI and railway/metro passenger volume: (a) relation with passenger volume; (b) Annual variation in the risk exposure rate.

in railroad passenger demand. Based on the ratio between annual SRMI and SMI (Fig. 15(b)), the year 2014 experienced relatively high levels of solar exposure, but heat-risk scenarios accounted for less than 5% of all cases. The decline in SRMI during this period is therefore more closely associated with thermal environmental conditions than with travel demand. Fig. 15(a) also indicates that the decline from 2020 to 2022 is related to reduced passenger volumes during the COVID-19 pandemic. Despite a substantial drop in total passenger volume from 2019 to 2020,

SRMI increased in 2020. When compared with the temporal distribution of WBGT in Fig. 4(a), the increase in the SRMI–SMI ratio corresponds to a higher frequency of heat-risk scenarios in 2020. This explains why UPS maintain a relatively high response performance even under significantly reduced passenger usage. Overall, these results indicate that the performance of the UPS in mitigating solar exposure during heatwave conditions is partly related to interannual changes in passenger usage levels. Travel demand and thermal conditions jointly shape the annual variation in SRMI.

3.4.2. Influence of activity-related factors

Three survey years (2012, 2017, and 2022) were selected to further examine the influence of travel demand at a micro temporal scale. Hourly mean SRMI values were calculated for each selected year. To account for differences between weekday and weekend travel patterns, Fig. 16 presents hourly mean SRMI as bar charts, with weekdays shown in green and weekends in yellow. Line plots indicate changes in total UPS trip counts by hour, with hollow circles for weekdays and filled diamonds for weekends.

Across all three years, no strong correspondence is observed between hourly SRMI and total UPS trip counts. Both weekday and weekend SRMI values exhibit a clear single-peak pattern. The effective response period of the UPS is mainly concentrated between 10:00 and 16:00. During weekday morning and evening commuting peaks, SRMI remains at zero, indicating that commuters do not benefit from the sun-shading function of the UPS. On weekends, UPS usage continues to increase during the afternoon, yet this does not prevent the continued decline in SRMI. Although weekend usage between 10:00 and 16:00 is consistently higher than on weekdays, weekend SRMI values do not exceed weekday levels. Taking 12:00 in 2022 as an example, weekday and weekend UPS usage reach similar levels, but SRMI is considerably higher on weekdays. One possible explanation is that the number of weekend days is lower than that of weekdays, and occasional rainy conditions on weekends can substantially reduce the average SRMI during these periods. Another possible explanation is differences in origin–destination distributions between weekday and weekend users, which may lead to variations in UPS response performance. Given that weekend SRMI exceeds weekday values at 13:00 in 2022, differences in the spatial distribution of travel demand may contribute to this result.

Table 3 presents the results of the univariate linear regression analyses examining the relationship between the cumulative hourly pedestrian volumes recorded during the three survey years and SRMI. The results indicate that variations in overall pedestrian volumes within the UPS have limited explanatory power for the temporal variation in SRMI. This finding suggests that the resilience performance of the UPS cannot be adequately explained solely by overall pedestrian volumes. Instead, understanding the influence of pedestrian volumes on the sun-exposure mitigation performance of the UPS requires consideration of both the dynamic spatial distribution of travel demand and its spatio-temporal coupling with sun-exposure risk distributions.

3.4.3. Influence of trip-distribution differences

To further test the hypotheses proposed in the previous section, this study used 12:00 as a representative time point and mapped the distribution of RMG for weekday and weekend trips in 2012, 2017, and 2022. As shown in Fig. 17, the colors of the connecting lines between street blocks and facilities indicate the levels of RMG. Connecting lines with warmer colors represent higher RMG values, while line widths reflect the number of users. For visual clarity, lines with RMG below 5% of the global maximum are omitted.

A comparison of high-benefit group distributions between weekdays and weekends reveals clear differences across years. In 2022, the spatial patterns of high-benefit groups are broadly similar between weekends (Fig. 17(f)) and weekdays (Fig. 17(e)). A key difference is that the weekday results in 2022 (Fig. 17(e)) shows a denser presence of lines associated with medium- and low-RMG groups. This indicates a more even distribution of RMG. The higher SRMI observed on weekdays during this period in Fig. 16(c) is therefore associated with a greater proportion of medium- and low-RMG groups. In contrast, both 2012 and 2017 exhibit pronounced differences between weekday and weekend patterns in terms of RMGs and travel volumes, which may contribute to the observed differences in overall SRMI.

Further comparison across years highlights substantial variation in the spatial distribution of high-RMG weekday trips. In 2012, high-RMG trips are mainly concentrated between the upper blocks (A1) and Tenjin Station (A2). In 2017, high-RMG flows shift toward routes between Tenjin Station (C2) and adjacent central blocks on the same side (C3–C5), while RMGs associated with trips originating from C1 decrease. By 2022, high-RMG flows are more concentrated along routes connecting E1 with central blocks (E3–E5) and lower blocks (E6).

Because climatic conditions, UPS usage levels, and travel distributions differ across years, the formation of weekday–weekend differences in SRMI cannot be fully explained by the distribution of RMG alone. Further investigation is required to clarify how the spatial distribution of travel demand and risk exposure areas jointly shapes inter-zone differences in RMG. This issue will be addressed in future work.

To further examine how the spatial distributions of travel demand and sun-exposure risk influence the overall SRMI, Fig. 18 presents a statistical analysis of the frequency of sun-exposed trips between underground entrances and exits across different RMG intervals. Trips without sun exposure are omitted from the analysis.

As shown in Fig. 18(a) and (f), the frequency of sun-exposed trips in the high-RMG intervals is higher on weekdays than on weekends in both 2012 and 2022. Consequently, the cumulative SRMI within the RMG range of 180–480 is consistently higher on weekdays than on weekends. Although the total UPS usage on weekdays in 2012 (21,879 people) is substantially higher than that on weekends (17,871 people), the difference in the frequency of sun-exposed trips is relatively small (Fig. 18(b)). As a result, the greater frequency of high-RMG trips leads to a higher overall SRMI on weekdays. A similar pattern is observed in 2022. Although UPS usage at 12:00 is comparable between weekdays and weekends, the lower proportion of sun-exposed trips and the greater concentration of trips within low-RMG intervals on weekends result in a substantially lower total SRMI than on weekdays (Fig. 18(f)).

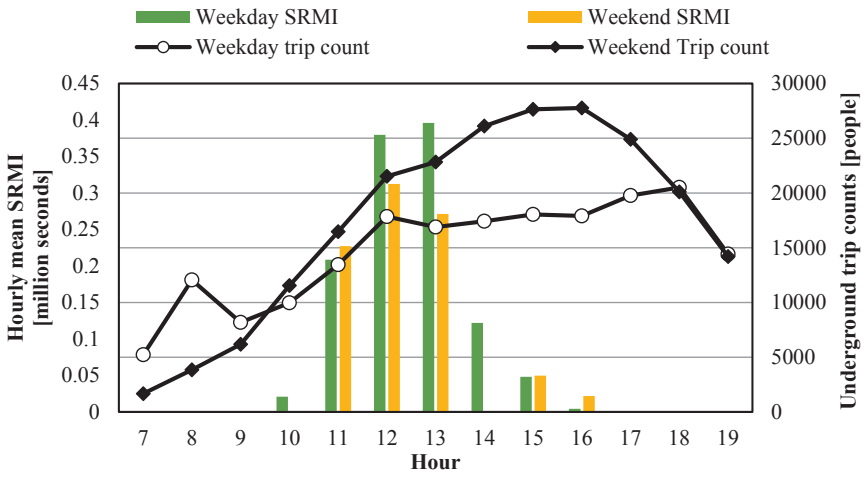
In contrast, the pattern in 2017 differs markedly. Despite similar UPS usage during the study period, the frequency of sun-exposed trips on weekends (6,718 people) is approximately 1.5 times higher than that on weekdays (Fig. 18(d)). Consequently, the frequency of sun-exposed trips is higher across all RMG intervals on weekends than on weekdays (Fig. 18(c)), resulting in a substantially higher total SRMI on weekends.

Further comparison across the three study years shows that the proportion of sun-exposed trips among all UPS trips was considerably higher in 2017 (22.4% on weekdays and 32.0% on weekends) than in 2012 (12.6% and 10.6%, respectively) and 2022 (13.9% and 10.4%, respectively). These differences reflect the combined influence of climatic conditions, urban morphology, and travel demand distribution. Therefore, the spatiotemporal variation in the system-level resilience performance of the UPS results from the interaction of multiple spatio-temporally heterogeneous factors, highlighting the complex system characteristics of three-dimensional pedestrian networks.

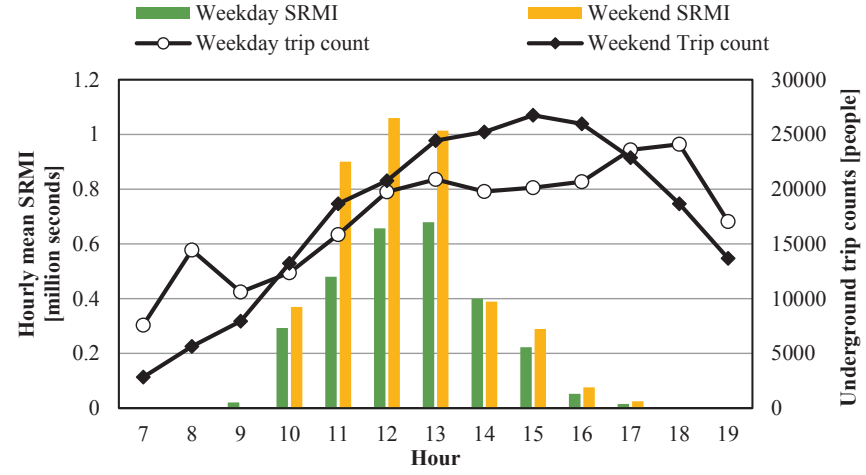
3.5. Sensitive analysis

A sensitivity analysis was conducted to evaluate the influence of pedestrian route-choice preferences on the simulation results. Because the route-choice model defines the UPS entrances and exits as trip origins and destinations, the utility weights associated with stair usage do not affect route selection on the ground-level pedestrian network. Therefore, the analysis focuses on examining how pedestrian preferences for shaded routes influence the simulated resilience performance of the UPS.

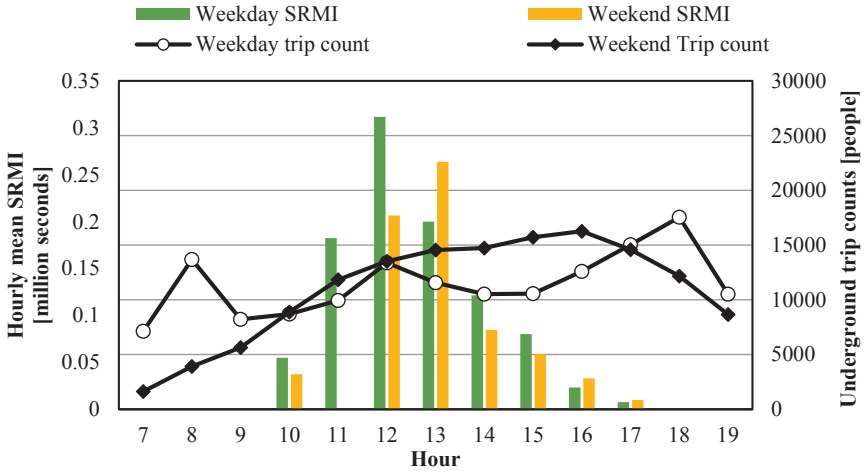
Following the route-choice model proposed by Melnikov et al.



(a)



(b)



(c)

Fig. 16. Comparison of hourly SRMI between weekdays and weekends in: (a) 2012; (b) 2017; and (c) 2022.

Table 3
Univariate linear regression analyses of the relationship between cumulative hourly pedestrian volumes and SRMI.

Year	Day type	Adjusted R^2	P value	N
2012	Weekday	−0.160	0.386	13
	Weekend	−0.160	0.387	13
2017	Weekday	0.090	0.156	13
	Weekend	−0.040	0.496	13
2022	Weekday	0.133	0.120	13
	Weekend	−0.910	0.978	13

(2022), pedestrian preference for shaded routes is represented by the parameter β. A value of β equal to 1 indicates that pedestrians exhibit no preference for shaded routes, whereas larger β values indicate a stronger tendency to choose more heavily shaded routes. In the sensitivity analysis, β was varied from 1.0 to 1.8, covering the most extreme scenario reported in the experimental study of Melnikov et al., with increments of 0.04. For each parameter value, pedestrian spatiotemporal distributions and the annual cumulative SRMI were simulated for each study year.

As shown in Fig. 19(a), the system-level cumulative SRMI gradually decreases as β increases, indicating that pedestrians with a stronger preference for shaded routes are more likely to adaptively adjust their travel routes to avoid sun exposure. Consequently, the relative contribution of the UPS to mitigating heat exposure is reduced. The cumulative SRMI relative deviations across all study years (Fig. 19(b)) further show that the overall variation in system performance resulting from changes in β ranges from − 4% to + 6%. These results indicate that the proposed framework is only weakly sensitive to the route-choice parameter, demonstrating the robustness and reliability of the simulation results.

4. Discussion

The proposed framework presents an advance in assessing the climate resilience contribution of the UPS to heat-exposure mitigation. By incorporating solar exposure during walking and waiting phases into pedestrian routing decisions, the model mathematically represents how users dynamically navigate through the ground-level street network. This allows for a more realistic estimation of the respective contributions of the UPS to the overall resilience of the pedestrian system under urban heatwaves. Furthermore, the integration of agent-based simulation under multiple temporal and spatial scenarios enables the model to reflect the variability and uncertainty of urban pedestrian responses to environmental conditions. This dynamic structure captures the complex interplay between climate disturbances, the pedestrian environment, and pedestrian decision-making processes, which is essential for understanding climate resilience as an emergent property of a complex system. In addition, the model jointly considers user-side travel demand and system-side service capacity, thereby providing a comprehensive framework for resilience assessment across different spatial units and time periods. The spatial visualization of performance indicators facilitates practical application in urban planning, allowing for evidence-based decision support in scenario evaluation, stakeholder decision-making, and the optimization of underground space design. Overall, the framework not only quantifies the climate resilience contribution of the UPS while incorporating more behaviourally realistic assumptions, but also provides a useful analytical framework urban design and management contexts where spatial equity and adaptive performance are increasingly important planning considerations. Compared with previous resilience assessment frameworks that primarily rely on static accessibility or risk indicators, the proposed framework explicitly incorporates pedestrian behavioural adaptation and spatiotemporal variability, thereby providing a more realistic representation of climate resilience in UPSs.

Despite the model's demonstrated contributions, several limitations should be acknowledged. Owing to data availability constraints, travel demand over the 13-year study period is estimated by interpolating pedestrian flows between the five-year interval pedestrian flow surveys. Although passenger-volume calibration partially accounts for the effects of climatic conditions and railway ridership on UPS demand, it does not fully capture the complexity and temporal variability of pedestrian flow distributions, which may influence long-term system performance. In addition, the framework does not explicitly consider the increase in UPS usage induced by extreme heat, which may further alter pedestrian travel patterns. Future research will integrate GPS trajectory data to improve the accuracy and reliability of travel demand estimation. Currently, solar exposure is the sole environmental variable integrated into the routing decision process, while other relevant climatic factors, such as ambient temperature, wind conditions, and humidity, remain unaccounted for. Incorporating these elements in future work would enhance the model's reliability in representing pedestrian behavioral responses to dynamic heat stress. Moreover, the current framework focuses exclusively on UPS users, thereby omitting pedestrians within the broader station area, whose exposure and routing decisions may also be shaped by underground accessibility. The assessment of UPSs' resilience contribution relies primarily on sunlight exposure duration of danger-level as the outcome indicator; future research should consider indices of other heat-risk levels and integrating comprehensive thermal indices, such as PET, while also accounting for heterogeneous comfort preferences across user groups. Additionally, the representation of origins and destinations by entrances and exits may oversimplify actual movement patterns within large facilities, potentially underestimating the spatial heterogeneity of beneficiary populations.

5. Conclusion

This study proposed a quantitative climate resilience assessment framework based on the path–risk relationship to evaluate the contribution of underground pedestrian systems (UPSs) to reducing pedestrians' sun-exposed heat risk during urban heatwaves. By integrating meteorological observations, urban morphology, pedestrian flow data, and multi-agent simulation, the framework quantified the resilience contribution of the Tenjin UPS under long-term summer climate scenarios. The major findings are summarized as follows.

(1) The proposed framework demonstrates that the UPS plays a significant role in mitigating pedestrians' exposure to direct solar radiation and dangerous heat conditions during summer heatwaves. The mitigation performance exhibits pronounced temporal heterogeneity across intraday, weekly, and interannual time scales, reflecting the combined influence of climatic variability and travel demand. The long-term simulations further reveal that prolonged heatwave periods substantially increase the resilience contribution of the UPS, whereas rainfall and cloud cover periodically interrupt its heat-risk mitigation response.
(2) The resilience benefits provided by the UPS are spatially heterogeneous. The proposed distribution-based indicators show that pedestrians travelling between different urban blocks receive substantially different levels of heat-risk mitigation, while the efficiency of underground entrances/exits also varies considerably across the network. Highly vulnerable street segments and several critical entrances/exits contribute disproportionately to the overall climate resilience of the pedestrian system, providing quantitative evidence for identifying priority locations for future infrastructure improvement and operational management.
(3) The temporal variation in UPS resilience is jointly controlled by environmental conditions and pedestrian travel demand. Air temperature, sunshine duration, and pedestrian volume collectively influence the temporal dynamics of the Sun-exposure Risk Mitigation Index (SRMI), and their effects vary across different temporal dimensions. The results further indicate that the

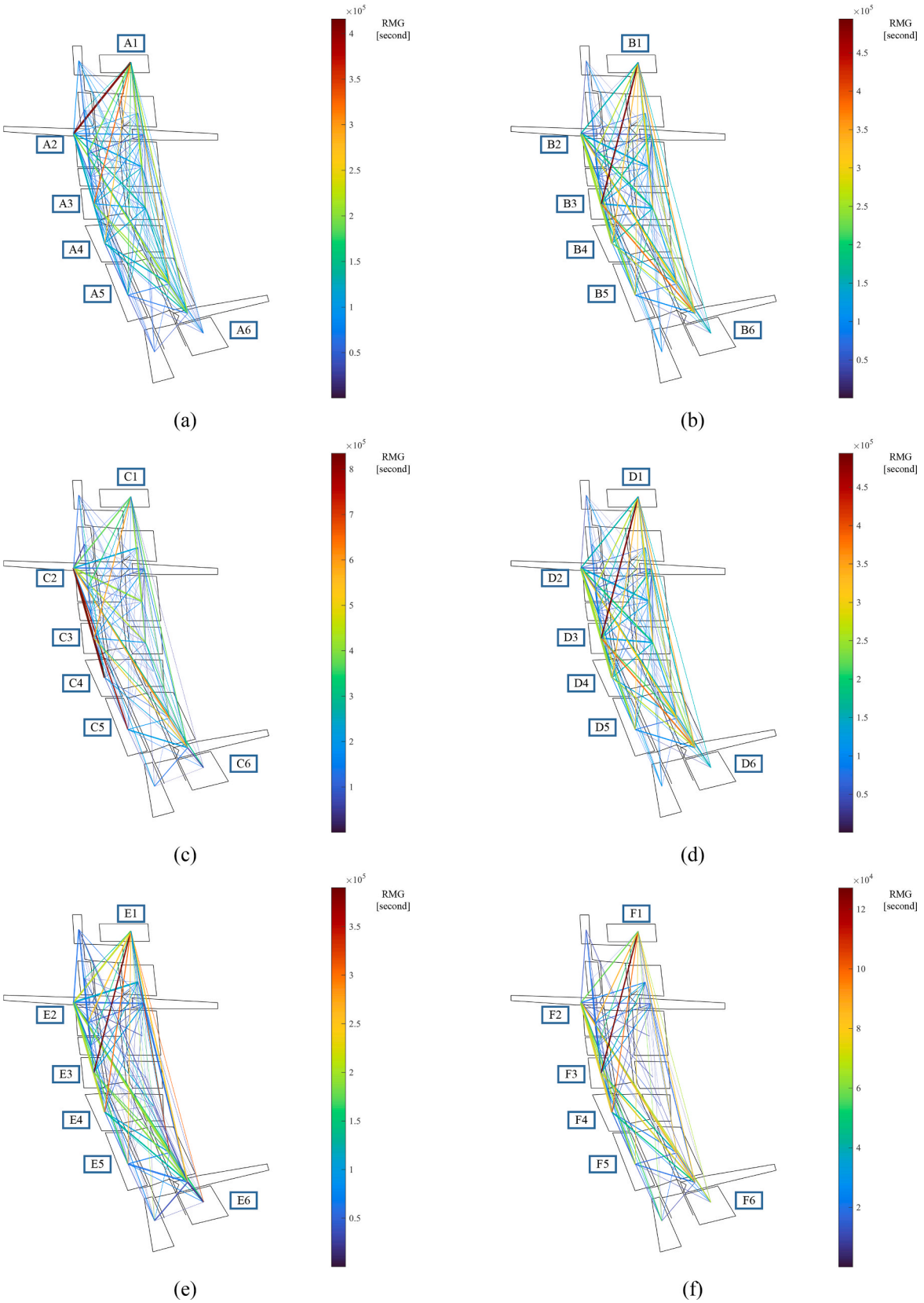
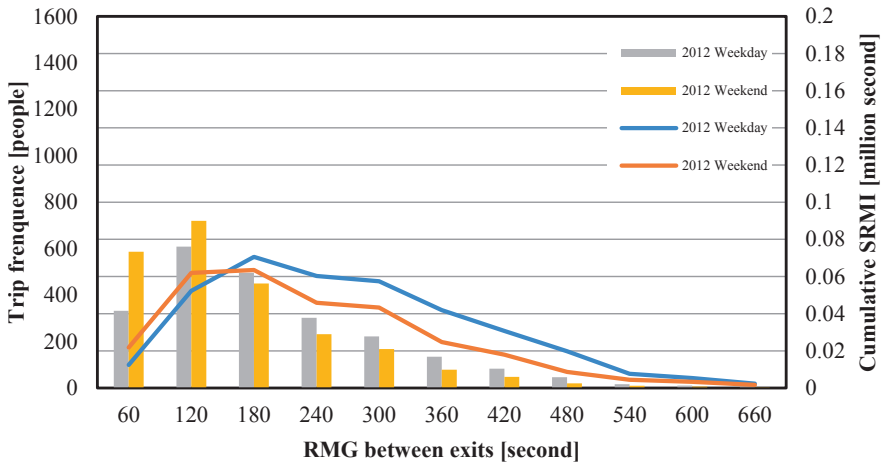
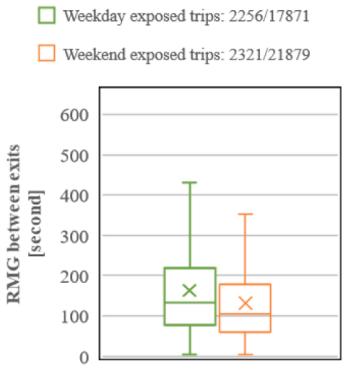


Fig. 17. Spatial distribution of RMG between urban blocks and facilities at 12:00 during (a) a weekday in 2012; (b) a weekend in 2012; (c) a weekday in 2017; (d) a weekend in 2017; (e) a weekday in 2022; and (f) a weekend in 2022.



(a)



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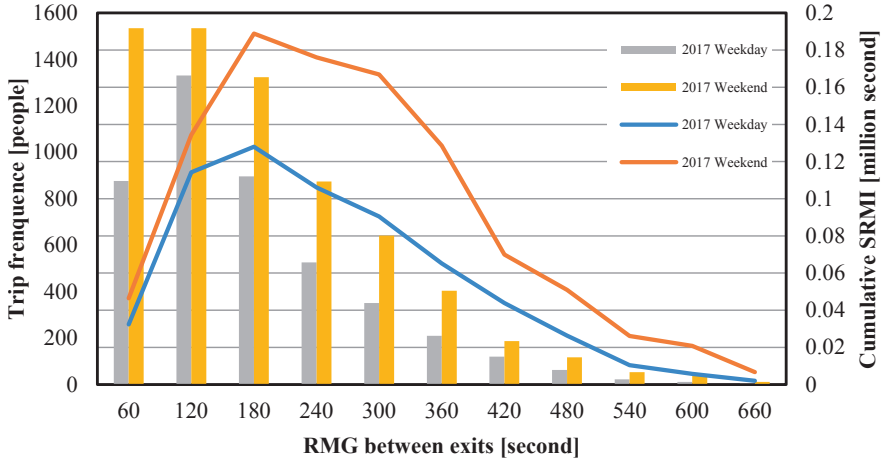

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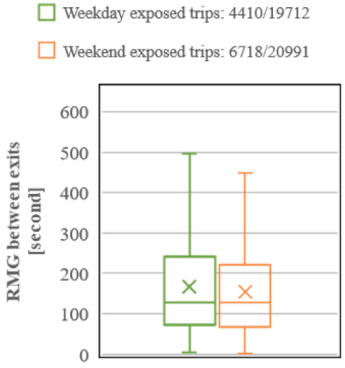

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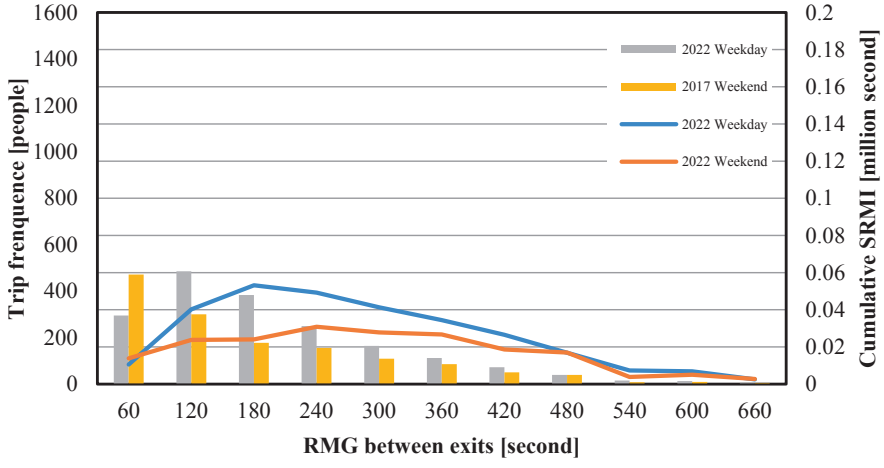

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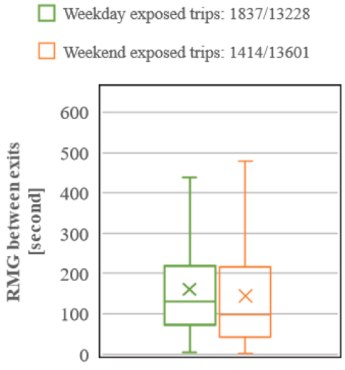

(f)

Fig. 18. Frequency distributions of sun-exposed trips and cumulative SRMI across different RMG intervals in 2012, 2017, and 2022.

seasonal peak of UPS resilience has become more concentrated and prolonged in recent years, suggesting an increasing role of UPSs under a warming climate.

(4) The comparison between sunshine exposure mitigation and extreme-heat-exposure mitigation reveals the climatic limitations of UPSs that rely solely on shading. During several extreme heatwave periods, even shaded outdoor environments reached dangerous WBGT levels, reducing the effectiveness of naturally ventilated underground pedestrian systems in mitigating heat risk. These findings indicate that future climate-adaptive

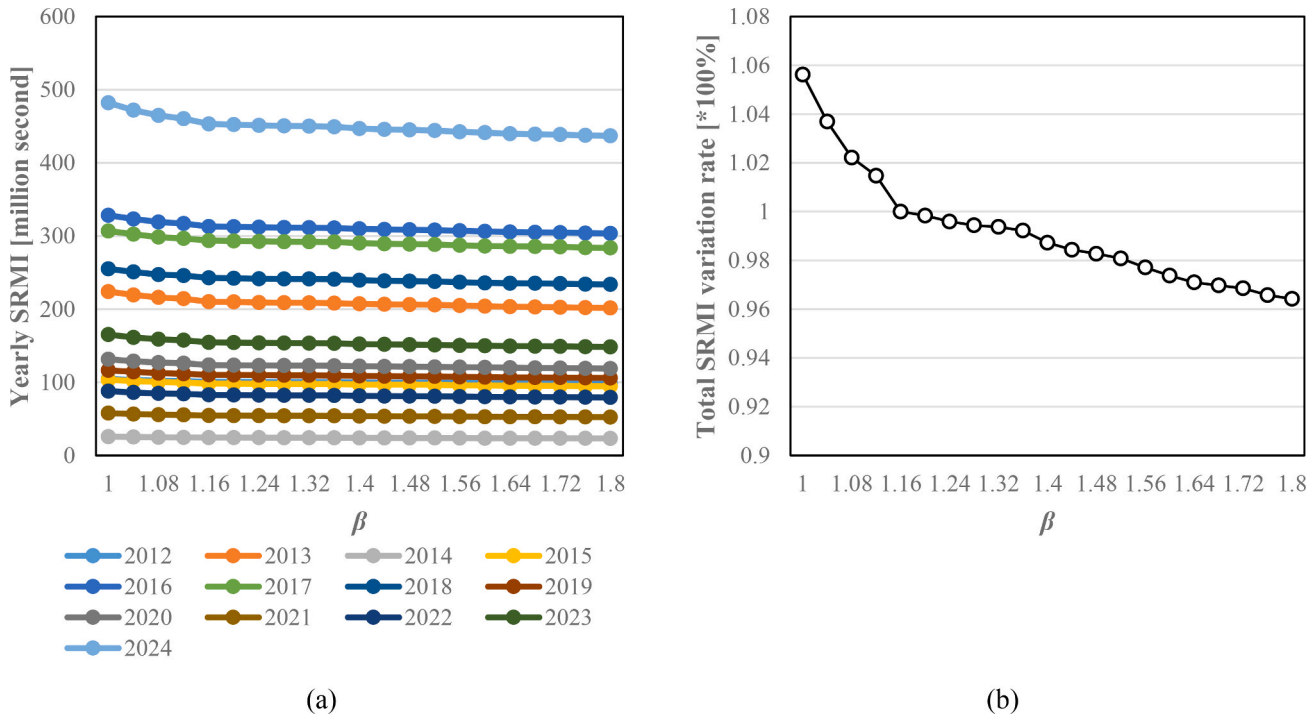

Fig. 19. Sensitivity of annual cumulative SRMI to the route-choice preference parameter (β): (a) cumulative SRMI under different β values; (b) relative deviations from the baseline scenario.

planning should not only optimize the spatial configuration of underground pedestrian networks but also strengthen indoor environmental control systems, particularly air-conditioning facilities, to maintain the resilience of UPSs under increasingly severe heatwave conditions.

Overall, the proposed framework provides a practical and quantitative tool for evaluating the climate resilience performance of underground pedestrian systems across multiple temporal and spatial scales. The findings provide useful guidance for identifying critical facilities, optimizing underground pedestrian networks, and supporting climate-adaptive urban planning and infrastructure renewal.

CRediT authorship contribution statement

Zongchao Gu: Writing – original draft, Visualization, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Toshihiro Osaragi:** Validation, Supervision, Project administration, Data curation, Conceptualization. **Jiawei Leng:** Supervision, Resources, Funding acquisition. **Haipeng Zhu:** Writing – original draft, Visualization, Resources, Methodology. **Nikolai Bobylev:** Writing – review & editing, Validation, Supervision.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT4 for improving language and readability. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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